

Series Representations of Fractional Brownian Motion

In this chapter, we present several special series representations of FBM. They are interesting in their own right, shed further light on FBM, and can be used in simulating FBM, for the study of its sample paths and other (e.g., statistical) tasks. We recall

- The Karhunen–Loève decomposition of stochastic processes (Section 8.1)

Except the case of BM, an explicit form of the basis functions (in terms of known special functions or transformations) in the Karhunen–Loève decomposition of FBM is currently not known. Two other interesting and explicit series representations of FBM are also considered:

- A wavelet-based expansion of FBM (Section 8.2)
- The Paley–Wiener representation of FBM (Section 8.3)

Since we will sometimes make reference to Chapter 6, we will parameterize FBM B_H with the self-similarity parameter $H \in (0, 1)$ as

$$B^\kappa, \quad \kappa = H - \frac{1}{2} \in \left(-\frac{1}{2}, \frac{1}{2}\right).$$

BM B_H with $H = 1/2$ is now associated with $\kappa = 0$, and is denoted B^0 . The material of Chapter 6 (more specifically, Sections 6.1–6.4) will be used only in Sections 8.2 and 8.3.

8.1 Karhunen–Loève Decomposition and FBM

We shall first recall the general Karhunen–Loève decomposition (Section 8.1.1), and then discuss it in the case of FBM (Section 8.1.3).

8.1.1 The Case of General Stochastic Processes

We recall here the Karhunen–Loève decomposition of general stochastic processes, following the formulation found in Ash and Gardner [52], Section 1.4. Let $\{X(t)\}_{t \in [0, a]}$ be a real-valued¹ stochastic process with $\mathbb{E}X(t) = 0$, $\mathbb{E}X(t)^2 < \infty$ and the covariance

$$K(t, s) = \mathbb{E}X(t)X(s), \quad s, t \in [0, a], \quad (8.1.1)$$

¹ Complex-valued processes can be considered as well.

which is assumed to be continuous on $[0, a] \times [0, a]$. Consider an integral operator

$$(Kf)(t) = \int_0^a K(t, s)f(s)ds, \quad t \in [0, a], \quad (8.1.2)$$

on $L^2[0, a]$. Since the operator is compact self-adjoint and by using the Hilbert–Schmidt theorem, there are eigenfunctions $\{\phi_n\}_{n \geq 1}$ and the corresponding eigenvalues $\{\lambda_n\}_{n \geq 1}$ satisfying

$$(K\phi_n)(t) = \int_0^a K(t, s)\phi_n(s)ds = \lambda_n\phi_n(t), \quad t \in [0, a] \quad (8.1.3)$$

(e.g., Gohberg, Goldberg, and Kaashoek [416], Section 5.1). The eigenfunctions are orthonormal in $L^2[0, a]$; that is,

$$\int_0^a \phi_m(s)\phi_n(s)ds = \begin{cases} 1, & \text{if } m = n, \\ 0, & \text{if } m \neq n. \end{cases} \quad (8.1.4)$$

Moreover, by Mercer's theorem,

$$K(t, s) = \sum_{n=1}^{\infty} \lambda_n \phi_n(t)\phi_n(s) \quad (8.1.5)$$

with the series converging absolutely and uniformly on $[0, a] \times [0, a]$ (e.g., Gohberg et al. [416], Section 5.2). The representation (8.1.5) leads to the following Karhunen–Loève decomposition (Ash and Gardner [52], Section 1.4.1; Riesz and Sz.-Nagy [850], Section 98).

Theorem 8.1.1 *In the setting described above,*

$$X(t) = \sum_{n=1}^{\infty} \sqrt{\lambda_n} \phi_n(t) Z_n, \quad t \in [0, a], \quad (8.1.6)$$

where

$$Z_n = \frac{1}{\sqrt{\lambda_n}} \int_0^a X(t)\phi_n(t)dt$$

are mean zero, variance one and uncorrelated random variables. The series (8.1.6) converges in $L^2(\Omega)$ uniformly in $t \in [0, a]$; that is,

$$\sup_{t \in [0, a]} \mathbb{E} \left| X(t) - \sum_{n=1}^N \sqrt{\lambda_n} \phi_n(t) Z_n \right|^2 \rightarrow 0, \quad (8.1.7)$$

as $N \rightarrow \infty$.

Note that changing a , changes the series expansion.

8.1.2 The Case of BM

The following example provides the well-known Karhunen–Loève decomposition of BM.

Example 8.1.2 (*Karhunen–Loève decomposition of BM*) For standard BM $B^0 = \{B^0(t)\}_{t \in [0, a]}$, the covariance is $K(t, s) = \min\{t, s\}$, $s, t \in [0, a]$. The eigenfunctions ϕ and eigenvalues λ satisfy the integral equation (8.1.3); that is,

$$\int_0^a \min\{t, s\} \phi(s) ds = \lambda \phi(t), \quad t \in [0, a], \quad (8.1.8)$$

that is,

$$\int_0^t s \phi(s) ds + t \int_t^a \phi(s) ds = \lambda \phi(t), \quad t \in [0, a]. \quad (8.1.9)$$

Differentiating (8.1.9) leads to

$$\int_t^a \phi(s) ds = \lambda \phi'(t) \quad (8.1.10)$$

and another differentiation to

$$-\phi(t) = \lambda \phi''(t). \quad (8.1.11)$$

This leads to the solution

$$\phi(t) = A \sin\left(\frac{t}{\sqrt{\lambda}}\right) + B \cos\left(\frac{t}{\sqrt{\lambda}}\right). \quad (8.1.12)$$

Setting $t = 0$ in (8.1.9) gives $\phi(0) = 0$ and hence $B = 0$. Setting $t = a$ in (8.1.10) leads to $\phi'(a) = 0$ and hence $\cos(a/\sqrt{\lambda}) = 0$ or $a/\sqrt{\lambda} = (2n-1)\pi/2$, $n \geq 1$. Thus, the eigenvalues and the normalized eigenfunctions are

$$\lambda_n = \frac{4a^2}{(2n-1)^2\pi^2}, \quad \phi_n(t) = \sqrt{\frac{2}{a}} \sin \frac{(2n-1)\pi t}{2a}, \quad (8.1.13)$$

for $n \geq 1$, where $A = \sqrt{2/a}$ ensures that $\int_0^a \phi_n(t)^2 dt = 1$. The Karhunen–Loève decomposition of BM is then

$$B^0(t) = \sqrt{\frac{2}{a}} \sum_{n=1}^{\infty} \frac{\sin((n-\frac{1}{2})\pi t/a)}{(n-\frac{1}{2})\pi/a} Z_n, \quad (8.1.14)$$

where Z_n , $n \geq 1$, are i.i.d. $\mathcal{N}(0, 1)$ random variables.

8.1.3 The Case of FBM

The standard FBM $B^\kappa = \{B^\kappa(t)\}_{t \in [0, a]}$, $\kappa \in (-1/2, 1/2)$, has the covariance $K(t, s) = \frac{1}{2}(t^{2\kappa+1} + s^{2\kappa+1} - |t-s|^{2\kappa+1})$, $s, t \in [0, a]$. The eigenfunctions ϕ_n and eigenvalues λ_n satisfy the integral equation

$$\frac{1}{2} \int_0^a (t^{2\kappa+1} + s^{2\kappa+1} - |t-s|^{2\kappa+1}) \phi_n(s) ds = \lambda_n \phi_n(t), \quad t \in [0, a]. \quad (8.1.15)$$

An explicit form for ϕ_n and λ_n satisfying (8.1.15) is currently not known. On the other hand, their asymptotic behavior is known as $n \rightarrow \infty$. As shown in Chigansky and Kleptsyna [236], Theorem 1.1, for $\kappa \neq 0$ and $a = 1$,

$$\lambda_n = \frac{\cos(\kappa\pi)\Gamma(2\kappa+2)}{(v_n)^{2\kappa+2}}, \quad n = 1, 2, \dots, \quad (8.1.16)$$

where

$$v_n = \left(n - \frac{1}{2}\right) - \frac{\kappa\pi}{2} + O(n^{-1}), \quad \text{as } n \rightarrow \infty. \quad (8.1.17)$$

Moreover,

$$\begin{aligned} \phi_n(t) &= \sqrt{2} \sin\left(v_n t + \frac{\kappa\pi}{4}\right) + \frac{\sqrt{2\kappa+2}}{\pi} \int_0^\infty \rho_0(u) \left((-1)^{n-1} e^{(t-1)v_n u} - u e^{-tv_n u}\right) du \\ &\quad + r_n(t) n^{-1}, \quad n = 1, 2, \dots, \end{aligned} \quad (8.1.18)$$

where the residual $r_n(t)$ is bounded by a constant depending only on κ , and

$$\rho_0(u) = \frac{\sin \theta_0(u)}{\gamma_0(u)} \exp\left\{\frac{1}{\pi} \int_0^\infty \frac{\theta_0(v)}{v+u} dv\right\}, \quad u > 0,$$

with

$$\begin{aligned} \theta_0(u) &= \arctan \frac{u^{-2\kappa-2} \sin(\kappa\pi)}{1 + u^{-2\kappa-1} \cos(\kappa\pi)}, \\ \gamma_0^2(u) &= (u + u^{-2\kappa-1} \cos(\kappa\pi))^2 + (u^{-2\kappa-1} \sin(\kappa\pi))^2. \end{aligned} \quad (8.1.19)$$

See also Chigansky and Kleptsyna [235]. This can be restated for general a using Exercise 8.1.

8.2 Wavelet Expansion of FBM

Meyer, Sellan, and Taqqu [716] obtained another interesting expansion of FBM based on wavelets. We shall outline their construction in this section, by recalling first the idea of wavelet bases (Section 8.2.1), then by considering the so-called fractional wavelets (Section 8.2.2) and finally by giving the wavelet-based expansion of FBM. It should also be noted from the start that the derived wavelet expansion is special in the sense that the underlying wavelet basis is adapted to the covariance structure of FBM in order to obtain an expansion with uncorrelated wavelet coefficients.

Warning: Recall that detail (wavelet) coefficients were used in estimation of the LRD parameter in Section 2.10.4. The detail coefficients of a process X were defined in (2.10.20)–(2.10.21) as $d(j, k) = \int_{\mathbb{R}} X(t) 2^{-j/2} \psi(2^{-j}t - k) dt$, where large j corresponds to large scales (low frequencies) and small (large negative) j corresponds to small scales (high frequencies). This choice of j is the convention used in the applied wavelet literature. In this section, we shall follow the mathematics convention where j is replaced by $-j$. In the mathematics convention, large j corresponds to small scales (high frequencies) and small (large negative) j corresponds to large scales (low frequencies). To avoid confusion, we shall write $d_{j,k}$ instead of $d(j, k)$.

8.2.1 Orthogonal Wavelet Bases

An orthonormal wavelet basis of $L^2(\mathbb{R})$ is a collection of functions

$$\phi(t - k), \quad k \in \mathbb{Z}; \quad 2^{j/2} \psi(2^j t - k), \quad j \geq 0, \quad k \in \mathbb{Z}, \quad (8.2.1)$$

with special properties. In particular, the functions are orthogonal, have unit $L^2(\mathbb{R})$ norms and allow expanding any $f \in L^2(\mathbb{R})$ as

$$f(t) = \sum_{k=-\infty}^{\infty} a_{0,k} \phi(t-k) + \sum_{j=0}^{\infty} \sum_{k=-\infty}^{\infty} d_{j,k} 2^{j/2} \psi(2^j t - k), \quad (8.2.2)$$

where

$$a_{0,k} = \int_{\mathbb{R}} f(u) \phi(u-k) du, \quad d_{j,k} = \int_{\mathbb{R}} f(u) 2^{j/2} \psi(2^j u - k) du. \quad (8.2.3)$$

The function ψ is known as a *wavelet* (mother wavelet), and ϕ as a *scaling function* (father wavelet). Note that $\phi(t-k)$, $k \in \mathbb{Z}$, are the translated copies of ϕ , and that $2^{j/2} \psi(2^j t - k)$, $j \geq 0$, $k \in \mathbb{Z}$, are the translated and scaled copies of ψ . The index k refers to location, and the index j (or rather the quantity 2^j) to scale or resolution. Note that large positive j corresponds to fine (small) scales while large negative j to coarse (large) scales. The coefficients $a_{0,k}$ and $d_{j,k}$ are known, respectively, as the *approximation* and *detail (wavelet) coefficients*.

Moreover, the functions ϕ and ψ can be related across different scales. More precisely, there are sequences $\{h_n\}_{n \in \mathbb{Z}}$ and $\{g_n\}_{n \in \mathbb{Z}}$ such that

$$2^{-1/2} \phi(2^{-1} t) = \sum_{n=-\infty}^{\infty} h_n \phi(t-n), \quad (8.2.4)$$

$$2^{-1/2} \psi(2^{-1} t) = \sum_{n=-\infty}^{\infty} g_n \phi(t-n). \quad (8.2.5)$$

The sequences $h = \{h_n\}_{n \in \mathbb{Z}}$ and $g = \{g_n\}_{n \in \mathbb{Z}}$ are known as *conjugate mirror filters* (CMFs). The filter h is low-pass² and satisfies

$$|\widehat{h}(w)|^2 + |\widehat{h}(w + \pi)|^2 = 2, \quad \widehat{h}(0) = \sqrt{2}, \quad (8.2.6)$$

where

$$\widehat{a}(w) = \sum_{n=-\infty}^{\infty} a_n e^{-inw}$$

denotes the Fourier transform of a sequence $a = \{a_n\}_{n \in \mathbb{Z}}$.³ The filter g is high-pass and related to h as

$$\widehat{g}(w) = e^{-iw} \overline{\widehat{h}(w + \pi)}, \quad (8.2.7)$$

where the bar indicates complex conjugation. See Exercise 8.2 for some other basic properties of CMFs h and g .

² A low-pass filter lets the low frequencies (large scales) through, and a high-pass filter lets the high frequencies (small scales) through.

³ See Appendix A.1.1. When consulting the wavelet literature, it is important to be aware of the conventions used for the Fourier transforms.

As the name suggests, a wavelet ψ is oscillating and wave-like. This is also reflected in the number of zero moments: ψ has Q zero moments when

$$\int_{\mathbb{R}} \psi(u) u^q du = 0, \quad q = 0, \dots, Q-1 \quad (8.2.8)$$

(cf. (2.10.19)). The larger the number of zero moments, the more oscillating ψ has to be. In contrast, a scaling function ϕ is usually such that

$$\widehat{\phi}(0) = \int_{\mathbb{R}} \phi(u) du = 1. \quad (8.2.9)$$

The relations (8.2.4)–(8.2.5) can be used to write (8.2.2) as follows: for any $J \in \mathbb{Z}$,

$$\begin{aligned} f(t) &= \sum_{k=-\infty}^{\infty} a_{J,k} 2^{J/2} \phi(2^J t - k) + \sum_{j=J}^{\infty} \sum_{k=-\infty}^{\infty} d_{j,k} 2^{j/2} \psi(2^j t - k), \\ &= \sum_{j=-\infty}^{\infty} \sum_{k=-\infty}^{\infty} d_{j,k} 2^{j/2} \psi(2^j t - k), \end{aligned} \quad (8.2.10)$$

where

$$a_{J,k} = \int_{\mathbb{R}} f(u) 2^{J/2} \phi(2^J u - k) du. \quad (8.2.11)$$

In the representation (8.2.10), the term $\sum_{k=-\infty}^{\infty} a_{J,k} 2^{J/2} \phi(2^J t - k)$ is thought as an approximation to f at scale J and the terms $\sum_{k=-\infty}^{\infty} d_{j,k} 2^{j/2} \psi(2^j t - k)$ as details at scales j . Moreover, if f is smooth, the approximation coefficients $\{a_{J,k}\}_{k \in \mathbb{Z}}$ approximate the function f as

$$2^{J/2} a_{J,k} \approx f\left(\frac{k}{2^J}\right). \quad (8.2.12)$$

The convergence rate in (8.2.12) is exponential in J for smooth f (Exercise 8.3).

Fast Wavelet Transform (FWT)

The CMFs $h = \{h_n\}_{n \in \mathbb{Z}}$ and $g = \{g_n\}_{n \in \mathbb{Z}}$ play a central role in the so-called *Fast Wavelet Transform* (FWT), which relates the approximation coefficients $a_j = \{a_{j,k}\}_{k \in \mathbb{Z}}$ and the detail coefficients $d_j = \{d_{j,k}\}_{k \in \mathbb{Z}}$ across different scales. It is used *at decomposition* to expand a function in wavelets and *at reconstruction* to obtain it back using the wavelet coefficients.

At decomposition, the FWT uses

$$a_{j-1} = \downarrow_2 (\bar{h} * a_j), \quad d_{j-1} = \downarrow_2 (\bar{g} * a_j), \quad (8.2.13)$$

where, for general sequences $x = \{x_n\}_{n \in \mathbb{Z}}$ and $y = \{y_n\}_{n \in \mathbb{Z}}$, $\bar{x}$ is the time reversion operation defined by $(\bar{x})_n = x_{-n}$, $\downarrow_2 x$ is the downsampling by 2 operation defined by $(\downarrow_2 x)_n = x_{2n}$ and $(x * y)_n = \sum_{k=-\infty}^{\infty} x_k y_{n-k}$ stands for the usual convolution. As illustration,

$$\downarrow_2 (\dots x_{-2} x_{-1} x_0 x_1 x_2 \dots) = (\dots x_{-2} x_0 x_2 \dots).$$

The FWT at decomposition thus allows obtaining the approximation and detail coefficients at coarser scale $j-1$ from the approximation coefficients at finer scale j .

At reconstruction, the FWT uses

$$a_j = h * (\uparrow_2 a_{j-1}) + g * (\uparrow_2 d_{j-1}). \quad (8.2.14)$$

Here, $\uparrow_2 x$ is the upsampling by 2 operation defined by $(\uparrow_2 x)_n = x_k$ if $n = 2k$, and $= 0$ if $n = 2k + 1$, for $k \in \mathbb{Z}$. As illustration,

$$\uparrow_2 (\dots x_{-2} x_{-1} x_0 x_1 x_2 \dots) = (\dots x_{-2} 0 x_{-1} 0 x_0 0 x_1 0 x_2 \dots).$$

The FWT at reconstruction thus allows obtaining the approximation coefficients at finer scale j from the approximation and details coefficients at coarser scale $j - 1$.

In practice, one usually works only with the CMFs h and g , and the approximation and detail coefficients a_j and d_j . If the data $f_k, k = 1, \dots, N$, are the function values $f(2^{-J}k)$ at scale J , the approximation coefficients would be taken as $2^{-J/2} f(2^{-J}k)$ in view of (8.2.12)⁴ and the FWT at decomposition would be used to compute the available approximation and detail coefficients at coarser scales. The FWT operation is of complexity $O(N)$ (with the sample size N above) and hence is very fast. Likewise, in simulations, one would start with, e.g., the approximation coefficients a_0 at scale 0 and then use the FWT at reconstruction to obtain the approximation coefficients a_J at finer scale J , which are then taken for the values of $2^{-J/2} f(2^{-J}k)$ according to the approximation (8.2.12).

The pair of wavelet ψ and scaling function ϕ (and the associated CMFs h and g) satisfying the properties above are referred to as *Multiresolution Analysis* (MRA).⁵

Example 8.2.1 (Haar MRA) An example is the Haar MRA with

$$\phi(t) = 1_{[0,1)}(t), \quad \psi(t) = 1_{[1/2,1)}(t) - 1_{[0,1/2)}(t) = \begin{cases} -1, & t \in [0, 1/2), \\ 1, & t \in [1/2, 1), \\ 0, & \text{otherwise} \end{cases} \quad (8.2.15)$$

and the CMFs

$$h_n = \begin{cases} 2^{-1/2}, & n = 0, 1, \\ 0, & \text{otherwise,} \end{cases} \quad g_n = \begin{cases} -2^{-1/2}, & n = 0, \\ 2^{-1/2}, & n = 1, \\ 0, & \text{otherwise} \end{cases} \quad (8.2.16)$$

(Exercise 8.4). As in this example, the functions ϕ and ψ with bounded supports are associated with finite CMFs h and g (that is, having only a finite number of nonzero elements).

Note also that ϕ and ψ in the Haar MRA are non-smooth, and ψ has only one zero moment $Q = 1$ in the sense of (8.2.8).⁶ Constructing smooth basis functions ϕ and ψ with compact supports and larger number of zero moments Q is, in fact, not an easy task. The celebrated Daubechies MRA provides examples of such bases for any desired number of zero moments $Q \geq 1$. When $Q = 1$, the Daubechies MRA is just the Haar MRA. When

⁴ This approximation is sometimes referred to as a “wavelet crime” in the wavelet literature.

⁵ Strictly speaking, the MRA refers to the sequence of approximation and detail subspaces of $L^2(\mathbb{R})$ induced by the wavelet and scaling functions.

⁶ The degree of smoothness and the number of zero moments are related. See, e.g., Mallat [675], Section 7.2.

$Q \geq 2$, the corresponding CMFs can be computed only numerically. See Daubechies [287], Meyer [715], Mallat [675] for more details.

Example 8.2.2 (*Daubechies MRA with $Q = 2$ zero moments*) When the number of zero moments $Q = 2$, the CMF h in the Daubechies MRA is given (up to 12 decimal places) by $h_0 = 0.482962913145$, $h_1 = 0.836516303738$, $h_2 = 0.224143868042$ and $h_3 = -0.129409522551$ (and other h_n 's being zero). The corresponding CMF g can be deduced from h by using (8.2.7). The functions ϕ and ψ can be approximated (computed numerically) by using the CMFs and the FWT at reconstruction as described above. More specifically, note that, for the function $f = \phi$,

$$a_{0,k} = \int_{\mathbb{R}} \phi(u)\phi(u-k)du = \begin{cases} 1, & k=0, \\ 0, & k \neq 0 \end{cases} \quad d_{j,k} = \int_{\mathbb{R}} \phi(u)2^{-j/2}\psi(2^{-j}u-k)du = 0,$$

for $j \geq 0$, $k \in \mathbb{Z}$, by orthonormality. Similarly, for the function $f = \psi$,

$$a_{0,k} = \int_{\mathbb{R}} \psi(u)\phi(u-k)du = 0, \quad d_{0,k} = \int_{\mathbb{R}} \psi(u)\psi(u-k)du = \begin{cases} 1, & k=0, \\ 0, & k \neq 0, \end{cases}$$

$$d_{j,k} = \int_{\mathbb{R}} \psi(u)2^{-j/2}\psi(2^{-j}u-k)du = 0,$$

for $j \geq 1$, $n \in \mathbb{Z}$. Having $a_0 = \{a_{0,k}\}_{k \in \mathbb{Z}}$ and $d_j = \{d_{j,k}\}_{k \in \mathbb{Z}}$, $j \geq 0$, the FWT at reconstruction given by (8.2.14) can be used to obtain the approximation coefficients $a_{J,k}$, $k \in \mathbb{Z}$, at finer and finer scales J . When normalized, $2^{J/2}a_{J,k}$ approximate $f(k/2^J)$ in view of (8.2.12). The plots of ϕ and ψ obtained through this procedure are given in Figure 8.1 for the Daubechies MRA with $Q = 2$ zero moments. A computationally more inclined and interested reader could try to implement the FWT procedure and to produce the plots in Figure 8.1.

Known examples of infinite MRAs are the Shannon MRA, the Battle–Lemarié MRA, the Meyer MRA and others. See, for example, Mallat [675].

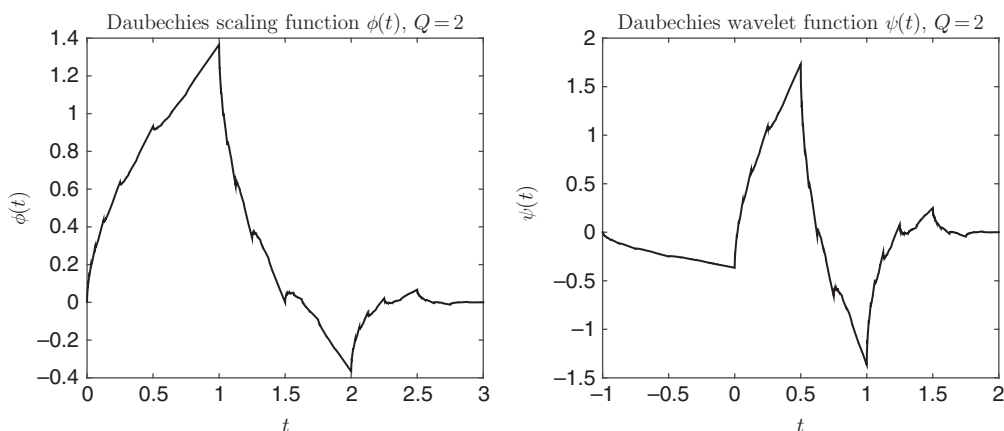


Figure 8.1 The Daubechies scaling and wavelet functions for $Q = 2$ zero moments.

The Meyer MRA

For technical convenience and following Meyer et al. [716], we shall work throughout with the Meyer MRA described as follows. Let $S(\mathbb{R})$ be the set of all infinitely differentiable functions $f(t)$ which satisfy

$$\lim_{|t| \rightarrow \infty} t^m \frac{d^n}{dt^n} f(t) = 0 \quad (8.2.17)$$

for any m and n ; that is, the tails of f and its all derivatives decrease to 0 faster than any polynomial.

In the Meyer MRA, a scaling function ϕ is such that $\phi \in S(\mathbb{R})$ or, equivalently, $\widehat{\phi} \in S(\mathbb{R})$ (Exercise 8.5), and satisfies:

$$\left\{ \begin{array}{l} \widehat{\phi}(x) \in [0, 1], \\ \widehat{\phi}(x) = 1 \quad \text{for } |x| \leq \frac{2\pi}{3}, \quad \widehat{\phi}(x) = 0 \quad \text{for } |x| \geq \frac{4\pi}{3}, \\ \text{and } \widehat{\phi}(x) \text{ decreasing on } (\frac{2\pi}{3}, \frac{4\pi}{3}), \\ \widehat{\phi}(-x) = \widehat{\phi}(x), \\ \sum_{n=-\infty}^{\infty} (\widehat{\phi}(\xi + 2\pi n))^2 = 1. \end{array} \right. \quad (8.2.18)$$

The last condition in (8.2.18) ensures that $\phi(t - n)$, $n \in \mathbb{Z}$, are orthonormal. There are many choices for the function ϕ (Meyer [715], Section VIII). The wavelet function is defined through⁷

$$\widehat{\psi}(x) = e^{ix/2} \left((\widehat{\phi}(x/2))^2 - (\widehat{\phi}(x))^2 \right)^{1/2}. \quad (8.2.19)$$

The wavelet ψ is such that $\psi \in S(\mathbb{R})$ and

$$\widehat{\psi}(x) = 0 \quad \text{if } |x| \leq 2\pi/3 \text{ or } |x| \geq 8\pi/3. \quad (8.2.20)$$

In particular, all the moments of ψ are zero since $\int_{\mathbb{R}} t^n \psi(t) dt = \widehat{\psi}^{(n)}(0)/i^n = 0$ for all $n \geq 0$. In contrast, $\widehat{\phi}(x) = 1$ in the neighborhood of $x = 0$. In the following section, we use the Meyer MRA unless stated otherwise.

8.2.2 Fractional Wavelets

We outline first the idea behind a wavelet-based decomposition of FBM, and then examine the resulting wavelet basis, which we call “fractional wavelets.” Recall from (6.2.29) that FBM can be represented as

$$B^\kappa(t) = \int_{\mathbb{R}} (I_-^\kappa 1_{[0,t)})(u) dB^0(u) =: \int_{\mathbb{R}} f_t(u) dB^0(u), \quad (8.2.21)$$

where

$$\mathbb{E} B^\kappa(t)^2 = \frac{c_{1,\kappa}^2}{\Gamma(\kappa + 1)^2} \quad (8.2.22)$$

⁷ Note that the convention for the Fourier transform used here is different from Meyer et al. [716].

and I_-^κ is a fractional integral or derivative of order κ . Since $f_t \in L^2(\mathbb{R})$, it can be expanded as in (8.2.2),

$$f_t(u) = \sum_{k=-\infty}^{\infty} a_{0,k}(t) \phi(u-k) + \sum_{j=0}^{\infty} \sum_{k=-\infty}^{\infty} d_{j,k}(t) 2^{j/2} \psi(2^j u - k), \quad (8.2.23)$$

where

$$a_{0,k}(t) = \int_{\mathbb{R}} f_t(u) \phi(u-k) du, \quad d_{j,k}(t) = \int_{\mathbb{R}} f_t(u) 2^{j/2} \psi(2^j u - k) du. \quad (8.2.24)$$

Substituting (8.2.23) into (8.2.21) yields the expansion

$$B^\kappa(t) = \sum_{k=-\infty}^{\infty} a_{0,k}(t) \eta_{0,k} + \sum_{j=0}^{\infty} \sum_{k=-\infty}^{\infty} d_{j,k}(t) \epsilon_{j,k}, \quad (8.2.25)$$

where

$$\eta_{0,k} = \int_{\mathbb{R}} \phi(u-k) dB^0(u), \quad \epsilon_{j,k} = \int_{\mathbb{R}} 2^{j/2} \psi(2^j u - k) dB^0(u). \quad (8.2.26)$$

By the orthonormality of the system $\phi(u-k)$, $2^{j/2} \psi(2^j u - k)$, $j \geq 0$, $k \in \mathbb{Z}$, the random variables $\eta_{0,k}$, $\epsilon_{j,k}$, $j \geq 0$, $k \in \mathbb{Z}$, are i.i.d. $\mathcal{N}(0, 1)$.

Equivalent Expression of Detail Coefficients $d_{j,k}(t)$

We shall rewrite next the decomposition (8.2.25) so that it has the form of a wavelet expansion (8.2.2).⁸ To do so, observe first that

$$\begin{aligned} d_{j,k}(t) &= \int_{\mathbb{R}} f_t(u) 2^{j/2} \psi(2^j u - k) du = \int_{\mathbb{R}} (I_-^\kappa 1_{[0,t)})(u) 2^{j/2} \psi(2^j u - k) du \\ &= \int_{\mathbb{R}} 1_{[0,t)}(s) (I_+^\kappa 2^{j/2} \psi(2^j \cdot - k))(s) ds = \int_{\mathbb{R}} 1_{[0,t)}(s) 2^{-j(\kappa-1/2)} (I_+^\kappa \psi)(2^j s - k) ds, \end{aligned} \quad (8.2.27)$$

where we used the integration by parts formula (6.1.27) and the fact that, e.g., for $\kappa > 0$,

$$\begin{aligned} (I_+^\kappa \psi(2^j \cdot - k))(s) &= \frac{1}{\Gamma(\kappa)} \int_{-\infty}^s \psi(2^j u - k) (s-u)^{\kappa-1} du \\ &= \frac{2^{-j\kappa}}{\Gamma(\kappa)} \int_{-\infty}^{2^j s - k} \psi(v) (2^j s - k - v)^{\kappa-1} dv = 2^{-j\kappa} (I_+^\kappa \psi)(2^j s - k) \end{aligned}$$

after making the change of variables $2^j u - k = v$. The relation (8.2.27) leads further to

$$\begin{aligned} d_{j,k}(t) &= 2^{-j(\kappa-1/2)} \int_0^\infty (I_+^\kappa \psi)(2^j s - k) ds - 2^{-j(\kappa-1/2)} \int_t^\infty (I_+^\kappa \psi)(2^j s - k) ds \\ &= 2^{-j(\kappa+1/2)} \int_{-k}^\infty (I_+^\kappa \psi)(u) du - 2^{-j(\kappa+1/2)} \int_{2^j t - k}^\infty (I_+^\kappa \psi)(u) du \end{aligned}$$

⁸ The reader not familiar with fractional integration, a topic introduced in Chapter 6, may jump directly to the definition (8.2.30) of the fractional wavelet $\widehat{\psi}^{(\alpha)}(x)$ and to the definition (8.2.34) of the fractional scaling function $\widehat{\phi}_{\Delta}^{(\alpha)}(x)$.

$$\begin{aligned}
&= 2^{-j(\kappa+1/2)}(I_+^1(I_+^\kappa\psi))(-k) - 2^{-j(\kappa+1/2)}(I_+^1(I_+^\kappa\psi))(2^j t - k) \\
&= 2^{-j(\kappa+1/2)}(I_+^{\kappa+1}\psi)(-k) - 2^{-j(\kappa+1/2)}(I_+^{\kappa+1}\psi)(2^j t - k),
\end{aligned} \tag{8.2.28}$$

where we made the change of variables $2^j s - k = u$ and also used the semigroup property (6.1.26). The presence of $I_+^{\kappa+1}\psi$ motivates the following definition.

Definition 8.2.3 A fractional wavelet of order $\alpha \in \mathbb{R}$ is defined as

$$\psi^{(\alpha)} = I_+^\alpha \psi \tag{8.2.29}$$

or through the Fourier transform as

$$\widehat{\psi}^{(\alpha)}(x) = \widehat{\psi}(x)(-ix)^{-\alpha} \tag{8.2.30}$$

(see Section 6.1.5).

Note that, for the Meyer MRA, $\widehat{\psi}^{(\alpha)} \in S(\mathbb{R})$ for all $\alpha \in \mathbb{R}$ since $\widehat{\psi} \in S(\mathbb{R})$ and $\widehat{\psi}(x) = 0$ in the neighborhood of $x = 0$ and $x = \infty$ (see (8.2.20)). Therefore, $\psi^{(\alpha)}$ is well-defined as the inverse of its Fourier transform and belongs to $S(\mathbb{R})$ (see Exercise 8.5).

In Section 8.2.4, we shall use (8.2.28) and hence the scaled and translated fractional wavelets to rewrite the detail coefficients $d_{j,k}(t)$ in (8.2.25). The approximation coefficients $a_{0,k}(t)$ in (8.2.25), on the other hand, cannot be dealt with in the same way as the detail coefficients $d_{j,k}(t)$ for two reasons. First, we are looking for a wavelet expansion where the coefficients $\eta_{0,k}$ approximate FBM $B^\kappa(k)$ (see (8.2.12)). This is certainly not possible in the expansion (8.2.25) since as k varies, $\eta_{0,k}$ are independent, whereas $B^\kappa(k)$ are dependent. Second, as indicated above, fractional wavelets are well-defined for the Meyer MRA for all $\alpha \in \mathbb{R}$ and hence can be used for $I_+^{\kappa+1}\psi$ in the representation (8.2.28). This is not the case for a scaling function ϕ . Since $\widehat{\phi}(x) = 1$ in the neighborhood of $x = 0$, $\widehat{\phi}(x)(-ix)^{-1-\kappa}$ is not in $L^2(\mathbb{R})$ and hence $I_+^{\kappa+1}\phi$ is not well-defined (at least, it cannot be defined through its Fourier transform). Both of these issues can be addressed simultaneously through the following modification.

Equivalent Expression for the Approximation Term

Informally, by arguing as for the detail coefficients (8.2.28) above, the approximation term in (8.2.25) is

$$\sum_{k=-\infty}^{\infty} a_{0,k}(t)\eta_{0,k} = \sum_{k=-\infty}^{\infty} (I_+^{\kappa+1}\phi)(-k)\eta_{0,k} - \sum_{k=-\infty}^{\infty} (I_+^{\kappa+1}\phi)(t-k)\eta_{0,k}.$$

Thus, up to a random shift and a change of sign, it is

$$Y_0(t) = \sum_{k=-\infty}^{\infty} (I_+^{\kappa+1}\phi)(t-k)\eta_{0,k}. \tag{8.2.31}$$

Since the function $I_+^{\kappa+1}\phi$ is not well-defined as argued above, the term (8.2.31) will be modified as follows. We proceed heuristically to arrive at expressions which are well-defined.

By using the relation $\widehat{f(\cdot - k)}(x) = e^{ixk} \widehat{f}(x)$, the “Fourier transform” of $Y_0(t)$ is

$$\widehat{Y}_0(x) = \widehat{I_+^{\kappa+1} \phi}(x) \sum_{k=-\infty}^{\infty} e^{ixk} \eta_{0,k} = \frac{\widehat{\phi}(x)}{(-ix)^{\kappa+1}} \overline{\widehat{\eta}_0(x)}.$$

Write this further as

$$\widehat{Y}_0(x) = \widehat{\phi}_{\Delta}^{(\kappa+1)}(x) \overline{\widehat{S}_0(x)}, \quad (8.2.32)$$

where

$$\widehat{\phi}_{\Delta}^{(\kappa+1)}(x) = \widehat{\phi}(x) \left(\frac{1 - e^{ix}}{-ix} \right)^{\kappa+1}, \quad \widehat{S}_0(x) = (1 - e^{-ix})^{-(\kappa+1)} \widehat{\eta}(x).$$

Note the correction term $(1 - e^{ix})^{\kappa+1}$ in $\widehat{\phi}_{\Delta}^{(\kappa+1)}(2^{-j}x)$ and its reciprocal in $\widehat{S}_0(x)$. Switching back to the time domain, Y_0 becomes

$$Y_0(t) = \sum_{k=-\infty}^{\infty} \phi_{\Delta}^{(\kappa+1)}(t - k) S_{0,k}. \quad (8.2.33)$$

This motivates the following definition.

Definition 8.2.4 A fractional scaling function of order $\alpha \in \mathbb{R}$ is defined through the Fourier transform as

$$\widehat{\phi}_{\Delta}^{(\alpha)}(x) = \widehat{\phi}(x) \left(\frac{1 - e^{ix}}{-ix} \right)^{\alpha}. \quad (8.2.34)$$

Lemma 8.2.5 For the Meyer MRA, the fractional scaling function $\phi_{\Delta}^{(\alpha)}$ is well-defined and belongs to $S(\mathbb{R})$ for all $\alpha \in \mathbb{R}$.

Proof It is enough to show that $\widehat{\phi}_{\Delta}^{(\alpha)}(x)$, given by the right-hand side of (8.2.30), belongs to $S(\mathbb{R})$ (see Exercise 8.5). We implicitly assume that the function $u(x) := (1 - e^{ix})/(-ix)$ is set to 1 at $x = 0$ so that it is continuous at this point. In particular, note that

$$u(x) = \frac{1 - e^{ix}}{-ix} = \frac{1}{ix} \sum_{m=1}^{\infty} \frac{(ix)^m}{m!} = \sum_{m=1}^{\infty} \frac{(ix)^{m-1}}{m!}$$

and that $u(x)$ is infinitely differentiable. Observe that

$$\frac{d^n \widehat{\phi}_{\Delta}^{(\alpha)}}{dx^n}(x) = \frac{d^n}{dx^n} \left((u(x))^{\alpha} \widehat{\phi}(x) \right) = \sum_{k=0}^n \binom{n}{k} \frac{d^k}{dx^k} (u(x))^{\alpha} \frac{d^{n-k} \widehat{\phi}}{dx^{n-k}}(x).$$

Since $\widehat{\phi} \in S(\mathbb{R})$ and $\widehat{\phi}(x) = 0$ for $|x| > 4\pi/3$ by (8.2.18), it is enough to show that $(u(x))^{\alpha}$ is infinitely differentiable on $[-4\pi/3, 4\pi/3]$. This follows from the facts that $u(x) \neq 0$ for $x \in [-4\pi/3, 4\pi/3]$ and, as indicated above, $u(x)$ is infinitely differentiable. $\square$

Let $\{X_k\}_{k \in \mathbb{Z}}$ be a generic sequence and let B be the backward shift operator as in Section 2.4.1, so that $(I - B)X_k = X_k - X_{k-1}$. Then the inverse operator $(I - B)^{-1}$ involves summation. In the Fourier world, $(I - B)$ is expressed as $(1 - e^{-ix})$ and $(I - B)^{-1}$ is expressed as $(1 - e^{-ix})^{-1}$.

We shall apply this here to $X_k = S_{0,k}$. This will allow us to identify the coefficients $S_0 = \{S_{0,k}\}_{k \in \mathbb{Z}}$ in (8.2.33). Express $\widehat{S}_0(x)$ by using two successive operations as

$$\widehat{S}_0(x) = (1 - e^{-ix})^{-(\kappa+1)} \widehat{\eta}_0(x) = (1 - e^{-ix})^{-1} (1 - e^{-ix})^{-\kappa} \widehat{\eta}_0(x) =: (1 - e^{-ix})^{-1} \xi_0(x).$$

The first operation $(1 - e^{-ix})^{-1} \xi_0(x)$ corresponds to taking a partial sum of the series ξ_0 . The second operation $(1 - e^{-ix})^{-\kappa} \widehat{\eta}_0(x)$, on the other hand, defines a Gaussian FARIMA(0, κ , 0) series (see Proposition 2.4.3). Thus,

$$S_{0,k} = \begin{cases} \sum_{m=1}^k \xi_{0,m}, & k \geq 1, \\ 0, & k = 0, \\ -\sum_{m=k+1}^0 \xi_{0,m}, & k \leq -1, \end{cases} \quad (8.2.35)$$

where

$$\xi_{0,k} = (I - B)^{-\kappa} \eta_{0,k} \quad (8.2.36)$$

is a Gaussian FARIMA(0, κ , 0) series. Note also that, in contrast to $\eta_{0,k}$ as discussed above, $S_{0,k}$ in (8.2.35) can now be thought as approximation coefficients to FBM. The expansion of FBM follows by using fractional scaling and wavelet functions and will be given in Section 8.2.4 below.

Biorthogonal Systems

We provide here some further insight and information on fractional wavelet and scaling functions. According to the discussion above, the system of functions

$$2^{J/2} \phi_{\Delta}^{(\alpha)}(2^J t - k), \quad 2^{j/2} \psi^{(\alpha)}(2^j t - k), \quad j \geq J, k \in \mathbb{Z}, \quad (8.2.37)$$

with $\alpha = \kappa + 1$ will be used in the wavelet expansion of FBM B^{κ} . Note that, unless $\alpha = 0$, this is not an orthogonal system of functions. In fact, the functions (8.2.37) are biorthogonal to the functions

$$2^{J/2} \phi_{\Delta}^{(-\alpha)}(2^J t - k), \quad 2^{j/2} \psi^{(-\alpha)}(2^j t - k), \quad j \geq J, k \in \mathbb{Z}, \quad (8.2.38)$$

in the sense that

$$\int_{\mathbb{R}} \left[2^{J/2} \phi_{\Delta}^{(\alpha)}(2^J u - k_1) \right] \left[2^{J/2} \phi_{\Delta}^{(-\alpha)}(2^J u - k_2) \right] du = \begin{cases} 1, & k_1 = k_2, \\ 0, & \text{otherwise,} \end{cases} \quad (8.2.39)$$

$$\int_{\mathbb{R}} \left[2^{j_1/2} \psi^{(\alpha)}(2^{j_1} u - k_1) \right] \left[2^{j_2/2} \psi^{(-\alpha)}(2^{j_2} u - k_2) \right] du = \begin{cases} 1, & k_1 = k_2, j_1 = j_2, \\ 0, & \text{otherwise} \end{cases} \quad (8.2.40)$$

(Exercise 8.6).

Moreover, the functions (8.2.37) and (8.2.38) form the so-called biorthogonal wavelet bases. This means, in particular, that the biorthogonal systems (8.2.37) and (8.2.38) are the so-called Riesz bases⁹ of $L^2(\mathbb{R})$, so that any $f \in L^2(\mathbb{R})$ can be expanded as

$$f(t) = \sum_{k=-\infty}^{\infty} c_{J,k} 2^{J/2} \phi_{\Delta}^{(\alpha)}(2^J t - k) + \sum_{j=J}^{\infty} \sum_{k=-\infty}^{\infty} e_{j,k} 2^{j/2} \psi^{(\alpha)}(2^j t - k), \quad (8.2.41)$$

⁹ A family of functions $\{g_l\}_{l \in \mathbb{Z}}$ is a Riesz basis of $L^2(\mathbb{R})$ if (i) there are constants $C_2 \geq C_1 > 0$ such that $C_1 (\sum_{l \in \mathbb{Z}} |a_l|^2)^{1/2} \leq \|\sum_{l \in \mathbb{Z}} a_l g_l\|_{L^2(\mathbb{R})} \leq C_2 (\sum_{l \in \mathbb{Z}} |a_l|^2)^{1/2}$ for all sequences $\{a_l\}_{l \in \mathbb{Z}} \in \ell^2(\mathbb{Z})$, and (ii) the linear span of $\{g_l\}_{l \in \mathbb{Z}}$ is dense in $L^2(\mathbb{R})$.

where

$$c_{J,k} = \int_{\mathbb{R}} f(u) 2^{J/2} \phi_{\Delta}^{(-\alpha)}(2^J u - k) du, \quad e_{j,k} = \int_{\mathbb{R}} f(u) 2^{j/2} \psi^{(-\alpha)}(2^j u - k) du. \quad (8.2.42)$$

See Theorem 1 in Meyer et al. [716]. Note that the indices are α and $(-\alpha)$ in (8.2.41) and (8.2.42), respectively.

8.2.3 Fractional Conjugate Mirror Filters

As with orthogonal wavelet bases, there is also a FWT (also known as Fast Biorthogonal Wavelet Transform) relating the approximation coefficients $c_j = \{c_{j,k}\}_{k \in \mathbb{Z}}$ and the detail coefficients $e_j = \{e_{j,k}\}_{k \in \mathbb{Z}}$ in (8.2.41) across different scales. At decomposition, the FWT uses

$$c_{j-1} = \downarrow_2 (\overline{h^{(\alpha)}} * c_j), \quad e_{j-1} = \downarrow_2 (\overline{g^{(\alpha)}} * c_j) \quad (8.2.43)$$

(cf. (8.2.13)), and at reconstruction, it uses

$$c_j = h^{(-\alpha)} * (\uparrow_2 c_{j-1}) + g^{(-\alpha)} * (\uparrow_2 e_{j-1}) \quad (8.2.44)$$

(cf. (8.2.14)), where $h^{(\alpha)} = \{h_n^{(\alpha)}\}_{n \in \mathbb{Z}}$ and $g^{(\alpha)} = \{g_n^{(\alpha)}\}_{n \in \mathbb{Z}}$ appear in

$$2^{-1/2} \phi_{\Delta}^{(\alpha)}(2^{-1}t) = \sum_{n=-\infty}^{\infty} h_n^{(\alpha)} \phi_{\Delta}^{(\alpha)}(t - n), \quad (8.2.45)$$

$$2^{-1/2} \psi^{(\alpha)}(2^{-1}t) = \sum_{n=-\infty}^{\infty} g_n^{(\alpha)} \phi_{\Delta}^{(\alpha)}(t - n) \quad (8.2.46)$$

(cf. (8.2.4) and (8.2.5)). See Section 7.4.1 in Mallat [675] for more details. Note that the indices are α and $(-\alpha)$ in (8.2.43) and (8.2.44), respectively. It can be checked (Exercise 8.7) that $h^{(\alpha)}$ and $g^{(\alpha)}$ are given as in the following definition.

Definition 8.2.6 The *fractional conjugate mirror filters (CMFs)* $h^{(\alpha)}$ and $g^{(\alpha)}$ for $\alpha \in \mathbb{R}$ are defined through their Fourier transform as

$$\widehat{h}^{(\alpha)}(x) = 2^{-\alpha} \widehat{h}(x) (1 + e^{-ix})^{\alpha}, \quad (8.2.47)$$

$$\widehat{g}^{(\alpha)}(x) = 2^{-\alpha} \widehat{g}(x) (1 - e^{-ix})^{-\alpha}, \quad (8.2.48)$$

where h and g are CMFs associated with an orthogonal wavelet basis.

Set informally

$$\widehat{p}^{(\alpha)}(x) = (1 + e^{-ix})^{\alpha}, \quad \widehat{q}^{(\alpha)}(x) = (1 - e^{-ix})^{-\alpha}, \quad (8.2.49)$$

so that $\widehat{h}^{(\alpha)}(x) = 2^{-\alpha} \widehat{h}(x) \widehat{p}^{(\alpha)}(x)$ and $\widehat{g}^{(\alpha)}(x) = 2^{-\alpha} \widehat{g}(x) \widehat{q}^{(\alpha)}(x)$. Note that it is not immediately clear why the fractional CMFs are well-defined. For example, note that, with $\alpha = \kappa + 1$ in the case of FBM, $(1 - e^{-ix})^{-\alpha}$ does not even belong to $L^2[-\pi, \pi]$.

One way to use the Fourier transform in the definition is through the number of zero moments of the underlying MRA. It is known that, under mild assumptions, a wavelet ψ has Q zero moments in the sense of (8.2.8) if and only if

$$\widehat{h}(x) = (1 + e^{-ix})^Q \widehat{h}_0(x), \quad \widehat{g}(x) = (1 - e^{-ix})^Q \widehat{g}_0(x) \quad (8.2.50)$$

with $\sup_{x \in [-\pi, \pi]} |\widehat{h}_0(x)| < \infty$ and $\sup_{x \in [-\pi, \pi]} |\widehat{g}_0(x)| < \infty$ (e.g., Mallat [675], Theorem 7.4). For example, for the Daubechies MRA, the filters h_0 and g_0 have also finite length, and can be found in Daubechies [287], Table 6.2 on p. 196.

Substituting (8.2.50) into (8.2.47)–(8.2.48) leads to

$$\widehat{h}^{(\alpha)}(x) = 2^{-\alpha} \widehat{h}_0(x) (1 + e^{-ix})^{\alpha+Q} = 2^{-\alpha} \widehat{h}_0(x) \widehat{p}^{(\alpha+Q)}(x), \quad (8.2.51)$$

$$\widehat{g}^{(\alpha)}(x) = 2^{-\alpha} \widehat{g}_0(x) (1 - e^{-ix})^{-\alpha+Q} = 2^{-\alpha} \widehat{g}_0(x) \widehat{q}^{(\alpha-Q)}(x). \quad (8.2.52)$$

There are several consequences of writing $\widehat{h}^{(\alpha)}$ and $\widehat{g}^{(\alpha)}$ as in (8.2.51)–(8.2.52). First, note that $\widehat{p}^{(\alpha+Q)}$ and $\widehat{q}^{(\alpha-Q)}$ are in $L^2[-\pi, \pi]$ when $-(\alpha + Q) < 1/2$ and $\alpha - Q < 1/2$, respectively. This is because $|\widehat{p}^{(\alpha+Q)}(x)|^2$ has to be integrable around $x = \pm\pi$, which requires $2(\alpha + Q) + 1 > 0$, and $|\widehat{q}^{(\alpha-Q)}(x)|^2$ has to be integrable around $x = 0$, which requires $2(-\alpha + Q) + 1 > 0$. Note that combining the two conditions leads to

$$-Q - 1/2 < \alpha < Q + 1/2.$$

The conditions hold for any α for large enough Q , showing that in this case, $h^{(\alpha)}$ and $g^{(\alpha)}$ are well defined through their Fourier transforms.

Second, the filters $p^{(\alpha+Q)}$ and $q^{(-\alpha+Q)}$ decay much faster than the corresponding filters $p^{(\alpha)}$ and $q^{(\alpha)}$ when Q is taken large. Indeed, $p^{(-a)} = \{p_n^{(-a)}\}_{n \in \mathbb{Z}}$ and $q^{(a)} = \{q_n^{(a)}\}_{n \in \mathbb{Z}}$ are well-defined through their Fourier transforms when $a < -1/2$. Their elements are the coefficients in the Taylor expansions of $(1+z)^{-a}$ and $(1-z)^{-a}$, respectively. Hence, in view of (2.4.2) and (2.4.4), the elements satisfy

$$(-1)^n p_n^{(-a)} = q_n^{(a)} = \prod_{k=1}^n \frac{a+k-1}{k} = \frac{\Gamma(n+a)}{\Gamma(n+1)\Gamma(a)} \sim \frac{n^{a-1}}{\Gamma(a)}, \quad (8.2.53)$$

as $n \rightarrow \infty$ (with $a \in \mathbb{R}$ not an integer). This implies that

$$p_n^{(\alpha+Q)} \sim (-1)^n \frac{n^{-1-Q-\alpha}}{\Gamma(-Q-\alpha)}, \quad q_n^{(\alpha-Q)} \sim \frac{n^{-1-Q+\alpha}}{\Gamma(-Q+\alpha)}, \quad (8.2.54)$$

as $n \rightarrow \infty$. The decay of the filters $p^{(\alpha+Q)}$ and $q^{(-\alpha+Q)}$ is important when truncating them in practice at some a priori chosen cutoff level ϵ .

Table 8.1, borrowed from Pipiras [813], provides the lengths of the truncated (normalized) fractional CMFs $2^\alpha h^{(\alpha)}$ and $2^\alpha g^{(\alpha)}$ at several cutoff levels ϵ for $\alpha = 1.25$ (with $\alpha = \kappa + 1$ corresponding to $\kappa = 0.25$) and the underlying Daubechies MRA with $Q = 1, 3, 6$ and 10 number of zero moments. The length of a truncated filter $2^\alpha h^{(\alpha)}$ for example, is defined as

$$\min\{N \geq 0 : |2^\alpha h_n^{(\alpha)}| \leq \epsilon, \text{ for all } n \geq N\}.$$

Note that high values of Q decrease significantly the length of the truncated fractional CMFs, especially in the case of $2^\alpha g^{(\alpha)}$. Note, however, as seen from the table entries for $\epsilon = 10^{-4}$ for example, taking Q larger and larger will not necessarily make the truncated fractional CMFs shorter. This happens because, for example, for the Daubechies MRA, the elements of the filters h_0 and g_0 in (8.2.50) and (8.2.51)–(8.2.52) tend to become larger as Q increases (see Daubechies [287], Table 6.2 on p. 196).

The discussion above points to an interesting and important role played by the number of zero moments of a wavelet-based expansion in the context of FBM. Recall from Section

Table 8.1 Lengths of truncated filters $2^\alpha h^{(\alpha)}$ and $2^\alpha g^{(\alpha)}$ at a cutoff ϵ with $\alpha = 1.25$ and the Daubechies MRA with Q vanishing moments.

Filters	Cutoff ϵ	Length of a truncated filter			
		$Q = 1$	$Q = 3$	$Q = 6$	$Q = 10$
$2^\alpha h^{(\alpha)}$	10^{-4}	17	17	21	28
	10^{-7}	123	48	36	38
	10^{-10}	1009	164	70	56
	10^{-15}	34769	1425	257	121
$2^\alpha g^{(\alpha)}$	10^{-4}	$\approx 4 \cdot 10^4$	38	23	28
	10^{-7}	$\approx 4 \cdot 10^8$	422	57	42
	10^{-10}	$\approx 4 \cdot 10^{12}$	5162	171	72
	10^{-15}	$\approx 4 \cdot 10^{19}$	$\approx 3 \cdot 10^5$	1220	206

2.10.4 that the number of zero moments was also important when using orthogonal wavelet basis in estimation of the LRD parameter.

8.2.4 Wavelet-Based Expansion and Simulation of FBM

By gathering the results and following the discussion above, we deduce next a wavelet-based expansion of FBM.

Theorem 8.2.7 Let $\kappa \in (-1/2, 1/2)$. FBM $B^\kappa = \{B^\kappa(t)\}_{t \in \mathbb{R}}$ in (8.2.21) admits the following wavelet expansion:

$$B^\kappa(t) = \sum_{k=-\infty}^{\infty} S_{0,k} \phi_{\Delta}^{(\kappa+1)}(t-k) + \sum_{j=0}^{\infty} \sum_{k=-\infty}^{\infty} 2^{-j(\kappa+1/2)} \epsilon_{j,k} \psi^{(\kappa+1)}(2^j t - k) - b_0, \quad (8.2.55)$$

where the convergence is uniform on compact intervals of $\mathbb{R}$ almost surely, $\phi_{\Delta}^{(\kappa+1)}$ and $\psi^{(\kappa+1)}$ are fractional scaling and wavelet functions appearing in Definitions 8.2.3 and 8.2.4, respectively, $\{\epsilon_{j,k}\}_{j \geq J, k \in \mathbb{Z}}$ are i.i.d. $\mathcal{N}(0, 1)$ random variables, $\{S_{0,k}\}_{k \in \mathbb{Z}}$ is a partial sum process of a Gaussian FARIMA(0, κ , 0) series defined in (8.2.35) (and independent of $\{\epsilon_{j,k}\}_{j \geq 0, k \in \mathbb{Z}}$), and b_0 is a random variable making the right-hand side of (8.2.55) zero at $t = 0$; that is,

$$b_0 = \sum_{k=-\infty}^{\infty} S_{0,k} \phi_{\Delta}^{(\kappa+1)}(-k) + \sum_{j=0}^{\infty} \sum_{k=-\infty}^{\infty} 2^{-j(\kappa+1/2)} \epsilon_{j,k} \psi^{(\kappa+1)}(-k). \quad (8.2.56)$$

Remark 8.2.8 In the case of BM with $\kappa = 0$ and when the underlying orthogonal wavelet basis is Haar, the expansion (8.2.61) is the celebrated midpoint displacement (Haar function) construction of BM due to Paul Lévy. See Section 2 in Meyer et al. [716] and also Section 2.3 in Karatzas and Shreve [549].

Sketch of Proof We shall use the fractional wavelets introduced in Section 8.2.2. In view of (8.2.25), (8.2.28) and (8.2.33), and by changing the signs of the random variables $\epsilon_{j,k}$ and $\eta_{0,k}$ in (8.2.25), we have

$$B^\kappa(t) = A_0(t) + D_0(t) - A_0(0) - D_0(0),$$

where

$$A_0(t) = \sum_{k=-\infty}^{\infty} S_{0,k} \phi_{\Delta}^{(\kappa+1)}(t-k), \quad D_0(t) = \sum_{j=0}^{\infty} \sum_{k=-\infty}^{\infty} 2^{-j(\kappa+1/2)} \epsilon_{j,k} \psi^{(\kappa+1)}(2^j t - k).$$

This holds rigorously and implies (8.2.55) as long as $D_0(t)$ is well-defined and converging uniformly on compact intervals as required, and

$$\sum_{k=-\infty}^{\infty} a_{0,k}(t) \eta_{0,k} = A_0(t) - A_0(0), \quad (8.2.57)$$

where $A_0(t)$ also converges uniformly on compact intervals. The coefficients $a_{0,k}(t)$ are defined in (8.2.24).

The equality (8.2.57) is proved in Lemma 9 and a discussion following it in Meyer et al. [716]. To show that $A_0(t)$ converges uniformly on compact intervals, note first that

$$|S_{0,k}| \leq C_0(1+|k|) \max_{m=-|k|+1, \dots, |k|} |\xi_{0,m}| \leq C(1+|k|) \sqrt{\log(2+|k|)} \quad (8.2.58)$$

for random constants C_0, C , where a FARIMA(0, κ , 0) sequence $\xi_{0,m}$ in (8.2.35) is bounded above by using Lemma 8.2.9 below. For any $u \in S(\mathbb{R})$, real $\delta \geq 0$ and slowly varying function L at infinity, one can choose constants $C_1, C_2 > 0$ such that

$$\begin{aligned} \sum_{k=-\infty}^{\infty} u(\tau-k)(1+|k|)^{\delta} L(1+|k|) &\leq C_1 \sum_{k=-\infty}^{\infty} (1+|\tau-k|)^{-q} (1+|k|)^{\delta} L(1+|k|) \\ &\leq C_2(1+|\tau|)^{\delta} L(1+|\tau|), \end{aligned} \quad (8.2.59)$$

by taking $q > 1 + \delta$ (Exercise 8.8). The bounds (8.2.58) and (8.2.59) with $u = \phi_{\Delta}^{(\kappa+1)}$, $\delta = 1$ and $L(1+|k|) = \sqrt{\log(2+|k|)}$ can be used to conclude that $|A_0(t)| \leq C(1+|t|)\sqrt{\log(2+|t|)}$ for a random constant C .

The uniform convergence of $D_0(t)$ on compact intervals can be proved similarly by using (8.2.59) (see the proof of Theorem 2 in Meyer et al. [716]). $\square$

The next auxiliary and useful lemma was used in the proof above.

Lemma 8.2.9 *Let $\{\varepsilon_n\}_{n \in \mathbb{Z}}$ be $\mathcal{N}(0, 1)$ random variables (which otherwise can be dependent). Then, there is a random constant $C(\omega)$ such that*

$$|\varepsilon_n(\omega)| \leq C(\omega) \sqrt{\log(2+|n|)}, \quad n \in \mathbb{Z}. \quad (8.2.60)$$

Proof Observe that, for $a > \sqrt{\log 2}$,

$$\sum_{n=3}^{\infty} \mathbb{P}(|\varepsilon_n| > a\sqrt{\log n}) \leq \sum_{n=3}^{\infty} \frac{2}{\sqrt{2\pi \log na}} e^{-\frac{a^2 \log n}{2}} \leq \sum_{n=3}^{\infty} n^{-\frac{a^2}{2}} < \infty,$$

where we used the inequality $\mathbb{P}(\mathcal{N}(0, 1) > x) \leq \frac{1}{\sqrt{2\pi}x} e^{-x^2/2}$ for $x > 0$. By the Borel–Cantelli Lemma, $|\varepsilon_n(\omega)| \leq a\sqrt{\log |n|} \leq a\sqrt{\log(2+|n|)}$ for $|n| > N(\omega)$ with a large

enough $N(\omega)$. The relation (8.2.60) then holds with $C(\omega) = a + \max\{\epsilon_n(\omega) : n = -N(\omega), \dots, N(\omega)\}/\sqrt{\log 2}$. $\square$

Simulation of FBM

Note that the expansion (8.2.55) of FBM is (8.2.41) with $J = 0$, $\alpha = \kappa + 1$ and $f(t) = B^\kappa(t) + b_0$. It can also be written as: for any $J \geq 0$,

$$B^\kappa(t) = \sum_{k=-\infty}^{\infty} 2^{-J(\kappa+1/2)} S_{J,k} \phi_{\Delta}^{(\kappa+1)}(2^J t - k) + \sum_{j=J}^{\infty} \sum_{k=-\infty}^{\infty} 2^{-j(\kappa+1/2)} \epsilon_{j,k} \psi^{(\kappa+1)}(2^j t - k) - b_0, \quad (8.2.61)$$

where

$$S_{J,k} = 2^{J(\kappa+3/2)} \int_{\mathbb{R}} (B^\kappa(u) + b_0) \phi_{\Delta}^{(-\kappa-1)}(2^J u - k) du. \quad (8.2.62)$$

by using the first relation in (8.2.42). The expansion (8.2.61) of FBM is now exactly (8.2.41) with $\alpha = \kappa + 1$ and

$$f(t) = B^\kappa(t) + b_0, \quad c_{J,k} = 2^{-J(\kappa+1/2)} S_{J,k} 2^{-J/2}, \quad e_{j,k} = 2^{-j(\kappa+1/2)} \epsilon_{j,k} 2^{-j/2}. \quad (8.2.63)$$

The expansion (8.2.61) could be used to simulate FBM. The variables $S_{J,k}$ are defined for $J \in \mathbb{Z}$ and $k \in \mathbb{Z}$. We can interpolate to get a step function $S_{J,[2^J t]}$ defined say for $t \in [0, 1]$. The relation (8.2.62) can be used to show that, for any $\epsilon \in (0, \kappa + 1/2)$,

$$\sup_{t \in [0, 1]} \left| 2^{-J(\kappa+1/2)} S_{J,[2^J t]} - (B^\kappa(t) + b_0) \right| \leq C 2^{-J(\kappa+1/2-\epsilon)} \quad (8.2.64)$$

almost surely, where the random variable C depends on κ , ϵ and the scaling function ϕ (Pipiras [813], Proposition 2.1). See also (8.2.12) and a related Exercise 8.3. The same result holds for the approximation of $B^\kappa(t)$ without b_0 if $S_{J,[2^J t]}$ is replaced by $S_{J,[2^J t]} - S_{J,0}$ (Pipiras [813], Corollary 2.1). Thus, suitably normalized and shifted coefficients $S_{J,[2^J t]}$ converge to FBM exponentially fast in J .

Substituting the coefficients c_j and e_j in (8.2.63) into (8.2.44) leads to the FWT at reconstruction given by

$$S_j = 2^{\kappa+1} h^{(-\kappa-1)} * (\uparrow_2 S_{j-1}) + 2^{\kappa+1} g^{(-\kappa-1)} * (\uparrow_2 \epsilon_{j-1}). \quad (8.2.65)$$

The FWT relation (8.2.65) could be used to generate the sequences $S_J = \{S_{J,k}\}_{k \in \mathbb{Z}}$ at finer and finer scales J from $S_0 = \{S_{0,k}\}_{k \in \mathbb{Z}}$ and $\epsilon_j = \{\epsilon_{j,k}\}_{k \in \mathbb{Z}}$, $j \geq 0$. Recall that $S_{0,k}$ is defined in (8.2.35) as a partial sum process of a FARIMA(0, κ , 0) sequence $\xi_{0,m}$. It can be shown that each S_J is a partial sum process of a Gaussian FARIMA(0, κ , 0) series as well (see Pipiras [813]). The relation (8.2.65) thus also provides a very fast (though approximate after truncating the fractional filters) method to generate partial sums of FARIMA(0, κ , 0) series. (See Pipiras [813] for further discussion.) The partial sums $S_J = \{S_{J,k}\}_{k \in \mathbb{Z}}$ of FARIMA(0, κ , 0) series, however, are special in the sense that they will converge to FBM almost surely as $J \rightarrow \infty$.

The left plot of Figure 8.2 presents the normalized and shifted approximations $S_{J,[2^J t]}$ to FBM $B^\kappa(t)$ on the interval $[0, 1]$ and for $\kappa = 0.25$. Linear interpolation instead of a step function is used. Thus, when $J = 0$, we get a straight line. In general, there are 2^J

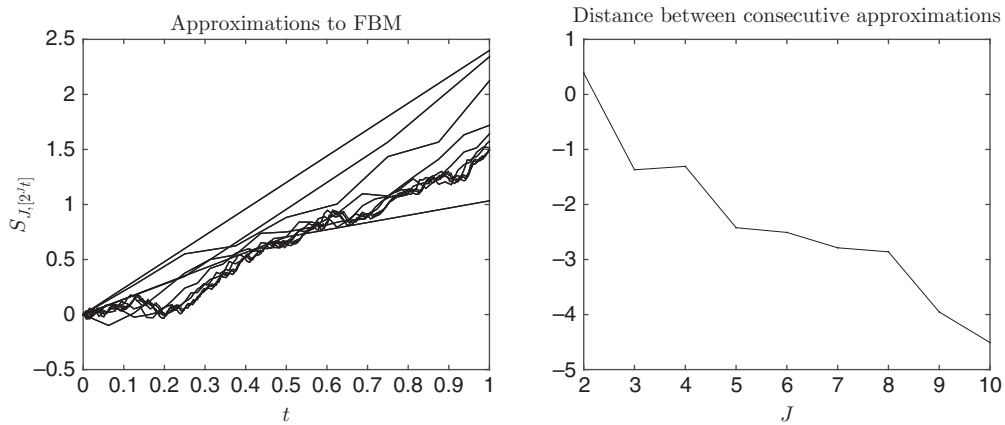


Figure 8.2 Left: Suitably normalized and shifted approximations $S_{J, [2^J t]}$ to FBM $B^\kappa(t)$ on the interval $[0, 1]$ and for $\kappa = 0.25$. Right: Uniform distances between the consecutive approximation on a log vertical scale.

connected line segments. The approximation improves as J increases. The right plot depicts the differences in uniform distance between the consecutive approximations on a log vertical scale. As in (8.2.64), the plot suggests that the convergence is exponentially fast in J .

8.3 Paley–Wiener Representation of FBM

An interesting expansion of FBM was obtained by Dzharidze and van Zanten [342]. The goal of this section is to sketch how this expansion is derived.

8.3.1 Complex-Valued FBM and its Representations

It is more convenient to work with complex-valued FBM defined as follows. Replace the Hermitian measure $\widehat{B}(dx)$ in the spectral representation (6.2.31) of FBM by the measure $\widetilde{B}(dx) = \widetilde{B}_1(dx) + i\widetilde{B}_2(dx)$, where $\widetilde{B}_1(dx)$ and $\widetilde{B}_2(dx)$ are independent Gaussian measures on $\mathbb{R}$ with control measures $dx/2$. A complex-valued FBM $\widetilde{B}^\kappa = \widetilde{B}_1^\kappa + i\widetilde{B}_2^\kappa$ is then defined as

$$\widetilde{B}^\kappa(t) = \frac{1}{c_{2,\kappa}} \int_{\mathbb{R}} \frac{e^{itx} - 1}{ix} |x|^{-\kappa} \widetilde{B}(dx), \quad (8.3.1)$$

where $c_{2,\kappa} = (2\pi)^{1/2} c_{1,\kappa} / \Gamma(\kappa + 1)$ with the constant $c_{1,\kappa}$ appearing in (6.2.31). It can be checked that the real part process $\sqrt{2}\Re \widetilde{B}^\kappa = \sqrt{2}\widetilde{B}_1^\kappa$ and the imaginary part process $\sqrt{2}\Im \widetilde{B}^\kappa = \sqrt{2}\widetilde{B}_2^\kappa$ define two (dependent) standard FBMs (Exercise 8.9).

Let

$$\mu(dx) = c_{2,\kappa}^{-2} |x|^{-2\kappa} dx, \quad (8.3.2)$$

$\widehat{1}_{[0,t)}(x) = \frac{e^{itx} - 1}{ix}$ be the Fourier transform of $1_{[0,t)}$ and consider the space

$$\mathcal{L}_a = \overline{\text{span}}\{\widehat{1}_{[0,t)} : t \in [0, a]\}, \quad (8.3.3)$$

where the closure of the linear span is in $L^2(\mathbb{R}, \mu) = \{f : \mathbb{R} \rightarrow \mathbb{C} : \int_{\mathbb{R}} |f(x)|^2 \mu(dx) < \infty\}$, equipped with the inner product $(f, g)_{L^2(\mathbb{R}, \mu)} = \int_{\mathbb{R}} f(x) \overline{g(x)} \mu(dx)$ of $L^2(\mathbb{R}, \mu)$. The space $\mathcal{L}_a$ is isomorphic to the Gaussian space

$$\widetilde{\mathcal{H}}_a = \overline{\text{span}}\{\widetilde{B}_H(t) : t \in [0, a]\}, \quad (8.3.4)$$

where the closure of the linear space is in $L^2(\Omega)$. Under the isomorphism, the function $\widehat{1}_{[0,t]}$ in $\mathcal{L}_a$ is associated with the random variable $\widetilde{B}^\kappa(t)$ in $\widetilde{\mathcal{H}}_a$. See Section 6.3.1 for the definitions of Gaussian spaces and isomorphisms.

If $\{\phi_n\}$ is any orthonormal basis in $\mathcal{L}_a$, then

$$\widehat{1}_{[0,t]}(x) = \sum_n c_n(t) \phi_n(x), \quad (8.3.5)$$

where

$$c_n(t) = \int_{\mathbb{R}} \widehat{1}_{[0,t]}(x) \overline{\phi_n(x)} \mu(dx). \quad (8.3.6)$$

Substituting (8.3.5) into (8.3.1) leads to the expansion

$$\widetilde{B}^\kappa(t) = \sum_n c_n(t) Z_n, \quad (8.3.7)$$

where $Z_n = \int_{\mathbb{R}} \phi_n(x) c_{2,\kappa}^{-1} |x|^{-\kappa} \widetilde{B}(dx)$ are uncorrelated complex-valued standard normal random variables (that is, satisfying $\mathbb{E}Z_n = 0$, $\mathbb{E}|Z_n|^2 = 1$ and $\mathbb{E}Z_n \overline{Z_m} = 0$). The convergence of the series (8.3.7) is in $L^2(\Omega)$ for each $t \in [0, a]$. Thus, the process defined by the right-hand side of (8.3.7) is a complex-valued FBM on $[0, a]$.

The expansion (8.3.7) holds for any orthonormal basis $\{\phi_n\}$ but the coefficients $c_n(t)$ in (8.3.6) do not necessarily have a closed form. Dzharidze and van Zanten [342] constructed a special orthonormal basis $\{\phi_n\}$ in $\mathcal{L}_a$ for which the coefficients $c_n(t)$ in (8.3.5)–(8.3.6) can be computed explicitly and which generalizes the Paley–Wiener expansion of BM (Paley and Wiener [788]). The latter is given by

$$\widetilde{B}^0(t) = \frac{1}{\sqrt{a}} \sum_{n=-\infty}^{\infty} \frac{e^{i2\pi nt/a} - 1}{i2\pi n/a} Z_n, \quad (8.3.8)$$

and defines a complex-valued BM. In the following section, we sketch the basis construction of Dzharidze and van Zanten [342].

8.3.2 Space $\mathcal{L}_a$ and its Orthonormal Basis

To work with the space $\mathcal{L}_a$ given in (8.3.3), it is convenient to introduce another space isomorphic to $\mathcal{L}_a$. Let $M^\kappa(t)$ be the fundamental martingale related to FBM $\widetilde{B}^\kappa$ through (see (6.3.49) and (6.3.50)):

$$M^\kappa(t) = \int_0^a m_t(u) d\widetilde{B}^\kappa(u), \quad (8.3.9)$$

where

$$m_t(u) = \frac{1}{\sigma_{2,\kappa}} u^{-\kappa} (I_{a-}^{-\kappa} 1_{[0,t]})(u) = \frac{u^{-\kappa} (t-u)^{-\kappa} 1_{[0,t]}(u)}{\sigma_{2,\kappa} \Gamma(1-\kappa)} \quad (8.3.10)$$

and $\sigma_{2,\kappa} = (2\kappa + 1)\Gamma(\kappa + 1)$. (The calculations below are based on second moments so that the complex-valued FBM in (8.3.9) will be viewed for simplicity as being real-valued.) Let

$$V(t) = \text{Var}(M^\kappa(t)) = \frac{\sigma_\kappa^2}{\sigma_{2,\kappa}^2(1-2\kappa)} t^{1-2\kappa} =: d_\kappa^2 t^{1-2\kappa} \quad (8.3.11)$$

be the quadratic variation appearing in (6.3.52). For later reference, note that

$$d_\kappa^2 = \frac{\Gamma(1-\kappa)}{(2\kappa+1)\Gamma(\kappa+1)\Gamma(2-2\kappa)} = \frac{\sqrt{\pi}}{(2\kappa+1)\Gamma(\kappa+1)\Gamma(3/2-\kappa)2^{1-2\kappa}}, \quad (8.3.12)$$

by using the identity $2^{1-2z}\sqrt{\pi}\Gamma(2z) = \Gamma(z)\Gamma(z+1/2)$. Note that $V(t) = t$ when $\kappa = 0$.

Inversion

The relation (8.3.9) can be inverted as done in (6.4.7)–(6.4.8) and other places in Sections 6.3 and 6.4. Informally, (8.3.9) yields

$$\begin{aligned} M^\kappa(t) &= \int_0^a 1_{[0,t)}(u) \dot{M}^\kappa(u) du = \int_0^a 1_{[0,t)}(u) m_t(u) \tilde{B}^\kappa(u) du \\ &= \int_0^a 1_{[0,t)}(u) \frac{1}{\sigma_{2,\kappa}} u^{-\kappa} (I_{a-}^{-\kappa} 1_{[0,t)})(u) \tilde{B}^\kappa(u) du \\ &= \int_0^a 1_{[0,t)}(s) \frac{1}{\sigma_{2,\kappa}} (I_{0+}^{-\kappa} u^{-\kappa} \tilde{B}^\kappa(u))(s) ds, \end{aligned}$$

that is, $\dot{M}^\kappa(s) = (\sigma_{2,\kappa})^{-1} (I_{0+}^{-\kappa} u^{-\kappa} \tilde{B}^\kappa(u))(s)$ or $\sigma_{2,\kappa} u^\kappa (I_{0+}^\kappa \dot{M}^\kappa)(u) = \tilde{B}^\kappa(u)$ or

$$\tilde{B}^\kappa(t) = \int_0^a x_t(s) dM^\kappa(s), \quad (8.3.13)$$

where

$$x_t(s) = \sigma_{2,\kappa} (I_{a-}^\kappa u^\kappa 1_{[0,t)}(u))(s) = \frac{\sigma_{2,\kappa}}{\Gamma(\kappa+1)} \left(t^\kappa (t-s)_+^\kappa - \kappa \int_s^t u^{\kappa-1} (u-s)_+^\kappa du \right), \quad (8.3.14)$$

by using (6.2.8).

Let now

$$\mathcal{K}_a = \overline{\text{span}}\{x_t : t \in [0, a]\}, \quad (8.3.15)$$

where the closure is in $L^2([0, a], dV)$. There is a natural isometry between $\mathcal{K}_a$ and $\tilde{\mathcal{H}}_a$ which associates x_t to $\tilde{B}^\kappa(t)$. Note that, under this isometry, $M^\kappa(t)$ is associated with $1_{[0,t)}$. So $1_{[0,t)} \in \mathcal{K}_a$ and hence $\mathcal{K}_a = L^2([0, a], dV)$.

The Three Isometric Hilbert Spaces

We now have three isometric Hilbert spaces associated with FBM:

- the linear Gaussian space $\tilde{\mathcal{H}}_a = \overline{\text{span}}\{\tilde{B}_H(t) : t \in [0, a]\}$,
- the frequency domain space $\mathcal{L}_a = \overline{\text{span}}\{\hat{1}_{[0,t)} : t \in [0, a]\}$, and
- the space of integration kernels $\mathcal{K}_a = \overline{\text{span}}\{x_t : t \in [0, a]\}$.

The isometries between $\tilde{\mathcal{H}}_a$ and $\mathcal{L}_a$, and between $\tilde{\mathcal{H}}_a$ and $\mathcal{K}_a$ are determined by associating $\tilde{B}^\kappa(t)$ to $\hat{1}_{[0,t]}$ and to x_t , respectively. These induce an isometry between the function spaces $\mathcal{K}_a$ and $\mathcal{L}_a$, which we denote by U :

$$U : \mathcal{K}_a \rightarrow \mathcal{L}_a. \quad (8.3.16)$$

Under the isometry $\mathcal{K}_a \rightarrow \tilde{\mathcal{H}}_a$, the indicator function $1_{[0,t]} \in \mathcal{K}_a$ with $t \in [0, a]$ is mapped to the random variable $M_t \in \tilde{\mathcal{H}}_a$. Under the spectral isometry $\tilde{\mathcal{H}}_a \rightarrow \mathcal{L}_a$, $M_t \in \tilde{\mathcal{H}}_a$ is mapped to the Fourier transform of the function m_t in (8.3.10); that is,

$$\hat{m}_t(\lambda) = \int_{\mathbb{R}} e^{i\lambda u} m_t(u) du \quad (8.3.17)$$

in $\mathcal{L}_a$. Thus,

$$(U1_{[0,t]})(\lambda) = \hat{m}_t(\lambda). \quad (8.3.18)$$

The next goal is to rewrite the right-hand side of (8.3.18) and to characterize the isometry U acting on general function $f \in \mathcal{K}_a$.

Expression of Uf and $U^{-1}\psi$

It can be verified (Exercise 8.10) that

$$\hat{m}_t(\lambda) = \begin{cases} \frac{\sqrt{\pi}}{(2\kappa+1)\Gamma(\kappa+1)} \left(\frac{t}{\lambda}\right)^{\frac{1}{2}-\kappa} e^{i\lambda t/2} J_{1/2-\kappa}\left(\frac{\lambda t}{2}\right), & \lambda \neq 0, \\ \frac{\sqrt{\pi}}{(2\kappa+1)\Gamma(\kappa+1)\Gamma(3/2-\kappa)2^{1-2\kappa}} t^{1-2\kappa}, & \lambda = 0, \end{cases} \quad (8.3.19)$$

where $J_{1/2-\kappa}$ is the Bessel function of the first kind of order $1/2 - \kappa$. Recall that for $\nu \neq -1, -2, \dots$, the Bessel function J_ν of the first kind of order ν can be defined on the region $\{z \in \mathbb{C} : |\arg(z)| < \pi\}$ as the absolutely convergent sum

$$J_\nu(z) = \sum_{n=0}^{\infty} \frac{(-1)^n (z/2)^{\nu+2n}}{\Gamma(n+1)\Gamma(\nu+n+1)}. \quad (8.3.20)$$

To derive the relation (8.3.19), the following Poisson's integral formula for the Bessel function is used: when $\Re(\nu) > -1/2$,

$$J_\nu(z) = \frac{(z/2)^\nu}{\Gamma(\nu+1/2)\Gamma(1/2)} \int_{-1}^1 (1-u^2)^{\nu-1/2} e^{izu} du \quad (8.3.21)$$

(Watson [999], Section 3, Eq. (3)). Note also that

$$\hat{m}_t(0) = \frac{\sqrt{\pi}}{(2\kappa+1)\Gamma(\kappa+1)\Gamma(3/2-\kappa)2^{1-2\kappa}} t^{1-2\kappa} = d_\kappa^2 t^{1-2\kappa} = V(t), \quad (8.3.22)$$

by using (8.3.12) and (8.3.11).

In view of (8.3.18) and (8.3.22), write

$$(U1_{[0,t]})(\lambda) = \int_0^t d\hat{m}_u(\lambda) = \int_0^t \frac{d\hat{m}_u(\lambda)}{d\hat{m}_u(0)} d\hat{m}_u(0) = \int_0^t \frac{d\hat{m}_u(\lambda)}{d\hat{m}_u(0)} dV(u). \quad (8.3.23)$$

By using the well-known formula for the Bessel functions

$$\frac{d}{dz}(z^\nu J_\nu(z)) = z^\nu J_{\nu-1}(z) \quad (8.3.24)$$

and the expression (8.3.19) of $\widehat{m}_t$, it can be checked that

$$\frac{d\widehat{m}_u(\lambda)}{d\widehat{m}_u(0)} = \frac{d\widehat{m}_u(\lambda)}{du} \left(\frac{d\widehat{m}_u(0)}{du} \right)^{-1} = \phi(\lambda u), \quad (8.3.25)$$

where

$$\phi(z) = \begin{cases} \Gamma(1/2 - \kappa) \left(\frac{z}{4}\right)^{\kappa+1/2} e^{iz/2} \left(J_{-1/2-\kappa}\left(\frac{z}{2}\right) + i J_{1/2-\kappa}\left(\frac{z}{2}\right) \right), & z \neq 0, \\ 1, & z = 0 \end{cases} \quad (8.3.26)$$

(Exercise 8.11). Thus,

$$(U 1_{[0,t)})(\lambda) = \int_0^a 1_{[0,t)}(u) \phi(\lambda u) dV(u) \quad (8.3.27)$$

for any $t \in [0, a]$. In fact, it can be shown (Dzhaparidze and van Zanten [342], Theorem 3.2) that, for $f \in \mathcal{K}_a$,

$$(Uf)(\lambda) = \int_0^a f(u) \phi(\lambda u) dV(u) \quad \text{a.e. } \mu(d\lambda) \quad (8.3.28)$$

and that

$$(U^{-1}\psi)(u) = \int_{\mathbb{R}} \psi(\lambda) \overline{\phi(u\lambda)} \mu(d\lambda) \quad \text{a.e. } dV(u), \quad (8.3.29)$$

for $\psi \in \mathcal{L}_a$ such that $\int_{\mathbb{R}} \|\phi(\cdot\lambda)\|_V |\psi(\lambda)| \mu(d\lambda) < \infty$.

Remark 8.3.1 Since

$$J_{1/2}(z) = \sqrt{\frac{2}{\pi z}} \sin(z), \quad J_{-1/2}(z) = \sqrt{\frac{2}{\pi z}} \cos(z), \quad z \neq 0 \ (z \in \mathbb{C}), \quad (8.3.30)$$

and $\Gamma(1/2) = \sqrt{\pi}$, it follows that $\phi(z) = e^{iz}$ in the standard BM case $\kappa = 0$. Thus, in this case, U is simply the Fourier transform. For general $\kappa \in (-1/2, 1/2)$, U can be viewed as a fractional version of the Fourier transform.

Remark 8.3.2 The following are some alternative representations of the quantities considered above in terms of fractional integrals and derivatives, provided the integrals are well defined:

$$\begin{aligned} \widehat{m}_t(\lambda) &= \frac{1}{\Gamma(\kappa)} I_{0+}^{1-\kappa} (u^{-\kappa} e^{iu\lambda})(t), \\ (Uf)(\lambda) &= \frac{1}{\Gamma(\kappa)} \mathcal{F}(u^{-\kappa} (I_{a-}^{-\kappa} f)(u))(\lambda), \\ (U^{-1}\psi)(t) &= \Gamma(\kappa) I_{a-}^{\kappa} (u^{\kappa} (\mathcal{F}^{-1}\psi)(u))(t), \end{aligned}$$

where $\mathcal{F}$ denotes the Fourier transform (Dzhaparidze and van Zanten [342], p. 623).

Reproducing Kernel on $\mathcal{L}_a$

An orthonormal basis of $\mathcal{L}_a = \overline{\text{span}}\{\widehat{1}_{[0,t]} : t \in [0, a]\}$ will be defined in terms of a reproducing kernel¹⁰ on $\mathcal{L}_a$. The reproducing kernel is a function $S_a(w, \lambda)$ defined on $\mathbb{R} \times \mathbb{R}$ and satisfying the following reproducing kernel relation: for $\psi \in \mathcal{L}_a$,

$$\int_{\mathbb{R}} \psi(\lambda) \overline{S_a(w, \lambda)} \mu(d\lambda) = \psi(w). \quad (8.3.31)$$

The reproducing kernel $S_a(w, \lambda)$ is defined as

$$S_a(w, \lambda) = \int_0^a \overline{\phi(uw)} \phi(u\lambda) dV(u), \quad (8.3.32)$$

where ϕ is given by (8.3.26) and V by (8.3.11). It satisfies the reproducing kernel relation (8.3.31) since, by using Fubini's theorem, (8.3.29) and (8.3.28),

$$\begin{aligned} \int_{\mathbb{R}} \psi(\lambda) \overline{S_a(w, \lambda)} \mu(d\lambda) &= \int_{\mathbb{R}} \psi(\lambda) \int_0^a \overline{\phi(uw)} \phi(u\lambda) dV(u) \mu(d\lambda) \\ &= \int_0^a \phi(uw) \left(\int_{\mathbb{R}} \psi(\lambda) \overline{\phi(u\lambda)} \mu(d\lambda) \right) dV(u) \\ &= \int_0^a \phi(uw) U^{-1} \psi(u) dV(u) \\ &= U(U^{-1} \psi)(w) = \psi(w). \end{aligned}$$

This is proved rigorously in Dzhpapridze and van Zanten [342], Theorem 4.1.

Note that the reproducing kernel can be written as

$$S_a(w, \lambda) = \int_0^a \overline{P(u, w)} P(u, \lambda) du, \quad (8.3.33)$$

where

$$P(u, \lambda) = \phi(u\lambda) \sqrt{\frac{dV(u)}{du}}. \quad (8.3.34)$$

By using the differential equations satisfied by $P(u, \lambda)$ and $P(u, \lambda)^* = e^{i\lambda u} \overline{P(u, \lambda)}$, Dzhpapridze and van Zanten [342] show that, for all $a > 0$ and $w, \lambda \in \mathbb{R}$,

$$i(\lambda - w) S_a(w, \lambda) = \overline{P(a, w)} P(a, \lambda) - \overline{P(a, w)^*} P(a, \lambda)^* \quad (8.3.35)$$

(Theorem 6.1 of Dzhpapridze and van Zanten [342]), and conclude that, for $w \neq \lambda$,

$$\begin{aligned} \frac{S_a(2w, 2\lambda)}{S_a(0, 0)} &= (1 - 2\kappa) \Gamma(1/2 - \kappa)^2 \left(\frac{a^2 w \lambda}{4} \right)^{\kappa+1/2} e^{ia(\lambda-w)} \times \\ &\quad \times \frac{J_{-1/2-\kappa}(aw) J_{1/2-\kappa}(a\lambda) - J_{1/2-\kappa}(aw) J_{-1/2-\kappa}(a\lambda)}{a(\lambda - w)} \end{aligned} \quad (8.3.36)$$

¹⁰ Note that the notion here is that of a reproducing kernel and not of a reproducing kernel Hilbert space as in Section 6.3.7.

and, for $w \in \mathbb{R}$,

$$\begin{aligned} \frac{S_a(2w, 2w)}{S_a(0, 0)} &= (1 - 2\kappa)\Gamma(1/2 - \kappa)^2 \left(\frac{aw}{2}\right)^{2\kappa+1} \times \\ &\times \left(J_{1/2-\kappa}(aw)^2 + \frac{2\kappa}{aw} J_{-1/2-\kappa}(aw) J_{1/2-\kappa}(aw) + J_{-1/2-\kappa}(aw)^2 \right) \end{aligned} \quad (8.3.37)$$

(Corollary 6.2 of Dzharidze and van Zanten [342]).

Orthonormal Basis in $\mathcal{L}_a$

The following result presents an orthonormal basis in $\mathcal{L}_a$ and the associated expansion formula (Dzharidze and van Zanten [342], Theorem 7.2). The zeros of the Bessel functions are well studied and understood, and can be computed in most mathematical software packages.

Let S_a be defined in (8.3.32), μ defined in (8.3.2) and V defined in (8.3.11).

Theorem 8.3.3 *Let $\dots < w_{-1} < w_0 = 0 < w_1 < \dots$ be the real zeros of $J_{1/2-\kappa}$ and, for $n \in \mathbb{Z}$, define the functions ψ_n on $\mathbb{R}$ by*

$$\psi_n(\lambda) = \frac{S_a(2w_n/a, \lambda)}{\|S_a(2w_n/a, \cdot)\|_\mu} \quad (8.3.38)$$

and set

$$\begin{aligned} \sigma(w_n)^{-2} &= S_a\left(\frac{2w_n}{a}, \frac{2w_n}{a}\right) \\ &= \begin{cases} (1 - 2\kappa)\Gamma(1/2 - \kappa)^2 \left(\frac{w_n}{2}\right)^{2\kappa+1} J_{-1/2-\kappa}(w_n)^2 V(a), & w_n \neq 0, \\ V(a), & w_n = 0, \end{cases} \end{aligned} \quad (8.3.39)$$

where V is defined in (8.3.11). The functions ψ_n form an orthonormal basis of $\mathcal{L}_a$, and every function $\psi \in \mathcal{L}_a$ can be expanded as

$$\psi(\lambda) = \sum_{n=-\infty}^{\infty} \sigma(w_n) \psi\left(\frac{2w_n}{a}\right) \psi_n(\lambda), \quad (8.3.40)$$

with the convergence taking place in $L^2(\mathbb{R}, \mu)$.

Observe that the functions ψ_n in (8.3.38) have unit norms by construction. They are also orthogonal. Indeed, for $n \neq m$,

$$\int_{\mathbb{R}} \psi_n(\lambda) \overline{\psi_m(\lambda)} \mu(d\lambda) = C_{n,m} \int_{\mathbb{R}} S_a\left(\frac{2w_n}{a}, \lambda\right) \overline{S_a\left(\frac{2w_m}{a}, \lambda\right)} \mu(d\lambda) = C_{n,m} S_a\left(\frac{2w_n}{a}, \frac{2w_m}{a}\right)$$

for a constant $C_{n,m} \neq 0$, by the reproducing kernel property (8.3.31). It remains to note that

$$S_a\left(\frac{2w_n}{a}, \frac{2w_m}{a}\right) = 0$$

by (8.3.36) since $J_{1/2-\kappa}(w_n) = 0$ and $J_{1/2-\kappa}(w_m) = 0$. The fact that $\{\psi_n\}$ is complete (and hence an orthonormal basis) is proved in Dzharapadze and van Zanten [342], Theorem 7.2. Finally, note also that the expansion (8.3.40) follows from

$$\|S_a(2w_n/a, \cdot)\|_\mu = \sigma(w_n)^{-1}$$

and

$$\int_{\mathbb{R}} \psi(\lambda) \overline{\psi_n(\lambda)} \mu(d\lambda) = \sigma(w_n) \psi\left(\frac{2w_n}{a}\right)$$

by the reproducing kernel property (8.3.31).

8.3.3 Expansion of FBM

The next result provides a Paley–Wiener expansion of FBM.

Theorem 8.3.4 *Let $\kappa \in (-1/2, 1/2)$ and $\dots < w_{-1} < w_0 = 0 < w_1 < \dots$ be the real zeros of $J_{1/2-\kappa}$. Let also Z_n , $n \in \mathbb{Z}$, be i.i.d. complex-valued Gaussian random variables with $\mathbb{E}Z_n = 0$ and $\mathbb{E}|Z_n|^2 = 1$, and let $\sigma(w_n)^2$ be given by (8.3.39). Then, the series*

$$\sum_{n \in \mathbb{Z}} \sigma(w_n) \frac{e^{i2w_n t/a} - 1}{i2w_n/a} Z_n \quad (8.3.41)$$

converges in $L^2(\Omega)$ for each $t \in [0, a]$ and defines a complex-valued FBM with index κ .

Proof We follow the general argument outlined in (8.3.5)–(8.3.7). The functions $\{\psi_n\}_{n \in \mathbb{Z}}$ given by (8.3.38) form an orthonormal basis of $\mathcal{L}_a$. As in (8.3.5), by using (8.3.40) with $\psi = \widehat{1}_{[0,t]}$, expand $\widehat{1}_{[0,t]} \in \mathcal{L}_a$ in the basis $\{\psi_n\}_{n \in \mathbb{Z}}$ as

$$\widehat{1}_{[0,t]}(x) = \sum_{n=-\infty}^{\infty} \sigma(w_n) \widehat{1}_{[0,t]}\left(\frac{2w_n}{a}\right) \psi_n(x) = \sum_{n=-\infty}^{\infty} \sigma(w_n) \frac{e^{i2w_n t/a} - 1}{i2w_n/a} \psi_n(x).$$

If $\widetilde{B}^\kappa$ is a complex-valued FBM, the general expansion (8.3.7) then yields

$$\widetilde{B}^\kappa(t) = \sum_{n=-\infty}^{\infty} \sigma(w_n) \frac{e^{i2w_n t/a} - 1}{i2w_n/a} Z_n. \quad \square$$

In the case $\kappa = 0$, we have $w_n = \pi n$ (see (8.3.30)) and $\sigma(w_n)^{-2} = a$, where the latter follows from (8.3.39) by using $V(a) = a$ when $\kappa = 0$ and the expression (8.3.30) for the Bessel function $J_{-1/2}$. Hence, when $\kappa = 0$, the expansion (8.3.41) indeed becomes the classical Paley–Wiener expansion (8.3.8) of BM.

Taking the real part of the complex-valued FBM and the Paley–Wiener expansion (8.3.41) leads to real-valued FBM and the following result.

Corollary 8.3.5 *Let X , $\{Y_n^{(1)}\}_{n \geq 1}$ and $\{Y_n^{(2)}\}_{n \geq 1}$ be i.i.d. $\mathcal{N}(0, 1)$ random variables. Let V be defined in (8.3.11) and $\sigma(w_n)^2$ be given by (8.3.39). Then, the series*

$$\frac{t}{\sqrt{V(a)}}X + \sum_{n=1}^{\infty} \frac{\sigma(w_n)}{\sqrt{2}} \frac{\sin(2w_nt/a)}{w_n/a} Y_n^{(1)} + \sum_{n=1}^{\infty} \frac{\sigma(w_n)}{\sqrt{2}} \frac{\cos(2w_nt/a) - 1}{w_n/a} Y_n^{(2)} \quad (8.3.42)$$

converges for each t in $L^2(\Omega)$ and defines (real-valued) standard FBM with index κ .

Proof Denote the complex-valued variables in (8.3.41) as $Z_n = Z_n^{(1)} + iZ_n^{(2)}$, where $Z_n^{(1)}, Z_n^{(2)}$ are i.i.d. $\mathcal{N}(0, 1/2)$ variables. The real part of the process (8.3.41) can then be written as

$$\sum_{n=-\infty}^{\infty} \sigma(w_n) \frac{\sin(2w_nt/a)}{2w_n/a} Z_n^{(1)} + \sum_{n=-\infty}^{\infty} \sigma(w_n) \frac{\cos(2w_nt/a) - 1}{2w_n/a} Z_n^{(2)}. \quad (8.3.43)$$

By observing that $w_{-n} = -w_n$ and $w_0 = 0$, and by writing $Z_n^{(1)} + Z_{-n}^{(1)} = Y_n^{(1)}$ and $Z_n^{(2)} - Z_{-n}^{(2)} = Y_n^{(2)}$ for i.i.d. $\mathcal{N}(0, 1)$ variables $Y_n^{(1)}, Y_n^{(2)}$, and $\sigma(w_0)tZ_n^{(0)} = tZ_n^{(0)}/\sqrt{V(a)} = tX/\sqrt{2V(a)}$ for $\mathcal{N}(0, 1)$ random variable X by using (8.3.39), the process (8.3.43) can be written as

$$\frac{t}{\sqrt{2V(a)}}X + \sum_{n=1}^{\infty} \sigma(w_n) \frac{\sin(2w_nt/a)}{2w_n/a} Y_n^{(1)} + \sum_{n=1}^{\infty} \sigma(w_n) \frac{\cos(2w_nt/a) - 1}{2w_n/a} Y_n^{(2)}. \quad (8.3.44)$$

It remains to recall (see the discussion following (8.3.1)) that a standard FBM is obtained by multiplying the real part of the complex-valued FBM by $\sqrt{2}$, leading to the expansion (8.3.42). $\square$

The convergence of the series (8.3.42) can be shown to be uniform in $t \in [0, a]$ with probability 1 by using the Itô–Nisio theorem (e.g., Kwapień and Woyczyński [586], Theorem 2.1.1). According to the theorem, since the summands of the series are independent across $n \geq 1$, for example the sequence

$$S_N(t) = \sum_{n=1}^N \frac{\sigma(w_n)}{\sqrt{2}} \frac{\sin(2w_nt/a)}{w_n/a} Y_n^{(1)}$$

converges uniformly on $[0, a]$ with probability 1 (that is, converges with probability 1 in $C[0, a]$ equipped with the uniform norm) if and only if S_N converges weakly in $C[0, a]$. For the weak convergence, since one already has the weak convergence of the finite-dimensional distributions (in fact, even the convergence in $L^2(\Omega)$), the weak convergence of the sequence S_N is equivalent to showing its tightness in $C[0, a]$. For a closely related sequence, this is proved in Theorem 4.5 of Dzhaparidze and van Zanten [341].

8.4 Exercises

Exercise 8.1 If $\phi_n = \phi_n^{(a)}$ and $\lambda_n = \lambda_n^{(a)}$ are, respectively, the eigenfunctions and eigenvalues for the integral equation (8.1.15) of FBM, show that

$$\phi_n^{(a)}(t) = \frac{1}{\sqrt{a}} \phi_n^{(1)}\left(\frac{t}{a}\right), \quad \lambda_n^{(a)} = a^{2\kappa+2} \lambda_n^{(1)}.$$

Hint: Use the $(2\kappa + 1)$ –self-similarity of FBM.

Exercise 8.2 Show that the CMFs h and g satisfy the following relations:

$$|\widehat{g}(w)|^2 + |\widehat{g}(w + \pi)|^2 = 2, \quad \widehat{g}(w)\overline{\widehat{h}(w)} + \widehat{g}(w + \pi)\overline{\widehat{h}(w + \pi)} = 0.$$

Hint: Use (8.2.7) and (8.2.6).

Exercise 8.3 Suppose that $f \in L^2(\mathbb{R})$ is Hölder with exponent β ; that is, $|f(t) - f(s)| \leq c|t - s|^\beta$ for some constant c and all $s, t \in \mathbb{R}$. Suppose also that the scaling function ϕ satisfies (8.2.9) and is such that $\int_{\mathbb{R}} |u|^\beta |\phi(u)| du < \infty$. Show that

$$\sup_{k \in \mathbb{Z}} \left| 2^{J/2} a_{J,k} - f\left(\frac{k}{2^J}\right) \right| \leq C 2^{-\beta J},$$

for some constant C . *Hint:* Write $2^{J/2} a_{J,k} - f(k/2^J) = 2^J \int_{\mathbb{R}} (f(u) - f(k/2^J)) \psi(u) du$.

Exercise 8.4 The Haar MRA is characterized by (8.2.15) and (8.2.16). Show that the Haar functions ϕ and ψ , and the CMFs h and g indeed satisfy the relations (8.2.4), (8.2.5) and (8.2.7).

Exercise 8.5 The space S consists of functions satisfying (8.2.17). Show that $f \in S$ if and only if $\widehat{f} \in S$.

Exercise 8.6 Prove the biorthogonality relations (8.2.39) and (8.2.40).

Exercise 8.7 By using (8.2.4) and (8.2.5), show that the fractional CMFs given in Definition 8.2.6 indeed satisfy the relations (8.2.45) and (8.2.45).

Exercise 8.8 Prove the second inequality in (8.2.59). *Hint:* For example, for $\tau > 0$ and integer for simplicity, split $\sum_{k=-\infty}^{\infty}$ into $\sum_{k=-\infty}^0 + \sum_{k=0}^{\tau/2} + \sum_{k=\tau/2}^{\tau} + \sum_{k=\tau}^{2\tau} + \sum_{k=2\tau}^{\infty}$ and consider each of the sums separately.

Exercise 8.9 Consider the complex-valued FBM defined in (8.3.1). Show that the real part process $\sqrt{2}\Re \widetilde{B}^\kappa = \sqrt{2}\widetilde{B}_1^\kappa$ and the imaginary part process $\sqrt{2}\Im \widetilde{B}^\kappa = \sqrt{2}\widetilde{B}_2^\kappa$ define two (dependent) standard FBMs.

Exercise 8.10 Verify the relation (8.3.19) by using (8.3.20).

Exercise 8.11 Verify the relation (8.3.25).

8.5 Bibliographical Notes

Section 8.1: The asymptotics of the eigenvalues and eigenvectors for FBM in (8.1.16) and (8.1.18) also appear in Bronski [188, 189], with applications to small ball problems. A more statistical application can be found in Li, Hu, Chen, and Zhang [623], though with a mistake indicated in van Zanten [972] and a further response in Li [622]. A related problem of the covariance structure of a discrete time FBM is investigated in Gupta and Joshi [441]. Finally,

we should also note the work of Maccone [659, 660], where the Karhunen–Loève expansion of a process related to FBM, and not that of FBM as claimed initially, was derived.

Section 8.2: As indicated a number of times in Section 8.2, the wavelet decomposition of FBM in Theorem 8.2.7 is due to Meyer et al. [716]. The idea of choosing a suitable (biorthogonal) wavelet basis so that detail coefficients of FBM and other processes become decorrelated can be traced at least to Benassi et al. [113], who used such wavelet decompositions to study sample properties of the processes (but who also did not specify the approximation term as in Theorem 8.2.7). Decompositions similar to that in Theorem 8.2.7 but based on splines appear in Blu and Unser [163], Unser and Blu [969]. Wavelet-based simulation of FBM was considered first in Abry and Sellan [7], and later revisited by Pipiras [813]. Rate optimality of the wavelet series approximations of FBM is studied in Ayache and Taqqu [62].

Similar wavelet-based decompositions and applications are considered for the Rosenblatt process in Pipiras [812], Abry and Pipiras [6], and for general stationary Gaussian processes in Didier and Pipiras [305, 306] (see also Unser and Blu [968]). Series representations of fractional Gaussian processes by trigonometric and Haar systems are studied in Ayache and Linde [61].

Section 8.3: This section is based on the work of Dzhaparidze and van Zanten [342]. A similar expansion of FBM, but involving the zeros of the Bessel functions $J_{-1/2-\kappa}$ and $J_{1/2-\kappa}$, was established in Dzhaparidze and van Zanten [341]. Dzhaparidze, van Zanten, and Zareba [344] provide a more general perspective on the expansion of FBM, based on a general theory valid for a large class of processes with stationary increments. Extensions to fractional Brownian sheet and optimality questions are studied in Dzhaparidze and van Zanten [343].

Other notes: A particular series expansion of FBM through the Lamperti transformation is considered in Baxevani and Podgórski [110]. Power series expansions for FBM are considered in Gilsing and Sottinen [389]. White and colored Gaussian noises as limits of sums of random dilations and translations of a single function appear in Gripenberg [432]. Functional quantization problems for FBM and other Gaussian processes are studied in Luschgy and Pagès [652, 653]. Construction of stationary self-similar generalized fields by random wavelet expansion can be found in Chi [234].